

An Exploration of Loss Functions in Multimodal Models and Their Impact on Downstream Performance

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ABSTRACT

Multimodal learning aims to develop models that can process and integrate information from different data modalities, such as images and text, within a unified representation space. Transformer-based models have become the standard approach in this field, enabling flexible and modality-agnostic learning frameworks. Despite rapid progress, multimodal models often differ across multiple design dimensions, including architecture, data, and training objectives, making it difficult to isolate which components drive model performance. While existing studies have investigated the role of data composition or architectural components, the controlled study of loss functions remains relatively underexplored, despite their importance in shaping a model’s representation space and influencing its downstream performance.

To address this gap, this thesis systematically studies how different combinations of loss functions affect representation quality as measured on downstream task performance. To isolate the effect of the loss, we keep the model architecture and all training parameters fixed and ablate only the losses. We design a dual-stream vision-language encoder based on the CLIP architecture that can flexibly accommodate any combination of four loss functions: two multimodal (contrastive image-text [CLIP] and image-text matching [ITM]) and two unimodal (image-only contrastive [SimCLR] and text-only masked language modeling [MLM]), with minimal architectural changes.

We evaluate the resulting models on zero-shot classification, image-text retrieval, and linear probe classification, and analyze representation similarity using Centered Kernel Alignment (CKA). We also perform an extensive ablation study on training dataset size to assess how the effectiveness of loss functions varies with data availability. Our results show that the CLIP contrastive loss alone suffices to learn effective multimodal representations, outperforming models trained with only SimCLR or ITM. While MLM improves performance on downstream tasks in combination with CLIP, SimCLR significantly degrades performance in our experiments. The ITM loss fails to learn suitable multimodal representations but shows a unique synergistic effect when paired with CLIP and MLM in our zero-shot classification experiments. In addition, we show that training CLIP with the unimodal losses improves data efficiency in the low-data regime. These findings establish the effectiveness of CLIP for learning visual representations with language supervision, as well as for learning multimodal representations which enable zero-shot capabilities via contrastive training. They also demonstrate that proper design and selection of loss functions is a critical component in building performant multimodal models.

Source code for this thesis is available at:

<https://github.com/philscho/MA-Thesis-Multimodal-Losses>

CONTENTS

1	INTRODUCTION	1
2	THEORETICAL BACKGROUND	5
2.1	Transformer architecture	5
2.1.1	BERT	6
2.1.2	Vision Transformer	7
2.2	Multimodal Transformers	8
2.2.1	Architecture: Single-Stream vs. Dual-Stream	9
2.2.2	Loss functions in multimodal learning	10
2.3	Models	11
2.3.1	BERT-inspired models	12
2.3.2	CLIP: Contrastive Language-Image Pre-training	14
2.4	Related work	15
3	EXPERIMENTAL DESIGN	19
3.1	Model architecture	19
3.2	Loss functions	21
3.2.1	Multimodal loss: Image-Text Contrastive (CLIP)	21
3.2.2	Multimodal loss: Image-Text Matching	22
3.2.3	Unimodal image loss: SimCLR	24
3.2.4	Unimodal text loss: Masked Language Modelling	26
3.3	Training data	28
3.4	Evaluation methods	29
3.4.1	Zero-shot image classification	29
3.4.2	Zero-shot image-text retrieval	30
3.4.3	Linear probing	30
3.4.4	Centered Kernel Alignment (CKA)	31
3.4.5	Test datasets	32
3.5	Implementation Details	33
3.5.1	Dataset Ablation	34

3.5.2	Evaluation Details	35
4	RESULTS	37
4.1	Zero-Shot Image Classification	37
4.2	Zero-Shot Image-Text Retrieval	43
4.3	Linear probing	45
4.4	Ablation of dataset size	49
4.4.1	Zero-shot classification	49
4.4.2	Zero-shot retrieval	51
4.4.3	Linear probing	51
4.5	Model Similarity Analysis	56
4.6	Discussion	59
5	CONCLUSION	65
5.1	Summary	65
5.2	Limitations and future research	66
	BIBLIOGRAPHY	69
	SUPPLEMENTARY TABLES	75
	ACKNOWLEDGEMENTS	79
	ERKLÄRUNG ZUR ABSCHLUSSARBEIT	81

1 INTRODUCTION

Our experience of the world is multi-modal. A modality can be defined as "a particular way of doing or experiencing something". [Cam25] Each of the five senses we experience reality with constitutes a different modality: seeing, hearing, smelling, tasting and touching. Data from each sense channel enters our nervous system, where a unified representation of the outer world is constructed that makes up our experience of the world. The quest to build machines that can replicate this process, learning representations that incorporate information from different modalities, lies at the heart of multimodal learning.

Early progress in modern multimodal learning has led to the development of models such as DALL-E [RPG⁺21] and Imagen [SCS⁺22], which are able to generate arbitrary images based on a text prompt, or reimagine images in different artistic styles. More recently, the integration of multimodal capabilities into large language models (LLMs) has led to the development of multimodal LLMs such as GPT-4V [Ope23] and Gemini. [ea25] These models represent powerful general-purpose AI assistants, capable of understanding and reasoning over visual and auditory inputs in addition to text.

A central challenge in multimodal learning is how to effectively combine different modalities into a shared or coordinated representation. Each modality has its own structural properties and inductive biases: images have spatial coherence and hierarchical visual patterns, while language is sequential and symbolic, governed by grammar and semantics. The introduction of the Transformer architecture has provided a general framework for modeling of diverse modalities. [VSP⁺] Originally developed for natural language processing, the Transformer's self-attention mechanism allows it to flexibly model dependencies within sequences of arbitrary input tokens, without relying on modality-specific inductive biases. [XZC] This flexibility has made it possible to design unified architectures that can process and integrate multiple modalities within a common architectural backbone. For this reason, multimodal models based on the Transformer architecture are at the core of modern multimodal learning.

These models typically use modality-specific encoders, such as Vision Transformers (ViT) [DBK⁺] for images and BERT-like models for text [DCLT], to map inputs into a shared embedding space, where cross-modal alignment and interaction can take place. Depending on the design, these models either fuse the modalities early using a single-stream encoder

or process them separately through dual-stream encoders, aligning their representations via specialized training objectives. [XZC]

As the field continues to evolve, new multimodal models are introduced at a rapid pace, often varying across multiple dimensions at once: from the scale and composition of training data to architectural choices and the loss functions used to optimize the models. [CZH⁺] This diversity of design can make it difficult to pinpoint which components are responsible for a model’s performance. While existing works have conducted ablation experiments to assess the influence of training data composition or specific encoder architectures [MGF⁺], controlled studies on the influence of loss functions on model performance and representation quality remain relatively underexplored. This constitutes a gap in the research literature, as the choice of loss functions used to train a model directly determines the kinds of features it learns, thereby shaping its representation space and influencing its performance on downstream tasks.

The aim of this thesis is to add to this line of investigation by conducting a controlled ablation study on different loss functions in multimodal learning. To properly isolate the effects of the loss function, we fix the model architecture and all training parameters, ablating only the choice of losses. We design a dual-stream vision-language encoder based on the CLIP [RKH⁺] architecture that can flexibly accommodate any combination of four loss functions, two multi-modal (contrastive image-text and image-text matching) and two unimodal (image-only contrastive and text-only masking), with minimal architectural changes.

After training, we test models on three downstream tasks: zero-shot image classification, zero-shot image-text retrieval and linear probe image classification. We also evaluate the effect on the learned representations using Centered Kernel Alignment (CKA) [KNLH]. In addition, we ablate the size of the pre-training dataset by training models on varying fractions of the dataset, studying how the effectiveness of each loss function varies with data availability.

Our key findings are:

- Contrastive Image-Text loss (CLIP) leads to the best results on all downstream tasks, establishing a performant baseline for zero-shot tasks and in linear probe classification.
- The addition of unimodal losses to CLIP has an auxiliary effect:
 - Masked Language Modelling (MLM) improves performance on zero-shot tasks, with slight benefits in linear probing
 - Image-only contrastive loss (SimCLR) significantly decreases performance on all downstream tasks

- Image-Text Matching (ITM) fails to learn suitable representations for zero-shot tasks and linear probing compared to CLIP, and degrades performance when used together with CLIP
- CLIP leads to more generalizable visual representations than self-supervision on images (SimCLR) alone
- SimCLR is more effective at learning representations from limited data than CLIP, but the addition of SimCLR as well as MLM to CLIP also improves its effectiveness in the low-data regime

This study proves the effectiveness of learning visual representations through language supervision with the CLIP loss, and demonstrates that the proper design and selection of loss functions in multimodal learning is essential to building performant models.

The rest of the thesis is structured as follows:

- Chapter 2 lays out the theoretical background necessary to understand modern multimodal learning. Starting with an introduction to the Transformer architecture as utilized for different modalities, in a uni- and multimodal setting, it then discusses two key components to multimodal models: architectural design (single-stream vs dual-stream), and types of loss functions. We present a variety of models using different combinations of these components, with an added emphasis on the CLIP architecture. The chapter ends with a discussion on related work.
- Chapter 3 details the experimental design of this study. It describes the specific model architecture, and explains the used loss functions, both in theory and in concrete implementation. It goes over the pre-training dataset and explains the evaluation methods used in this work. Lastly, it gives details on implementation for model training and evaluations.
- Chapter 4 presents and discusses the results of our experiments. It first goes through each task in order: zero-shot image classification, zero-shot image-text retrieval and linear probing. It then goes over the results of the dataset ablation experiments for each task. These results are supplemented with a short section on model similarity analysis using CKA, before the chapter concludes with a discussion of the results for each loss function.
- Chapter 5 concludes the thesis by giving a summary of our work, before discussing its limitations and directions for future research.

1 Introduction

Source code for this thesis is available at:

<https://github.com/philscho/MA-Thesis-Multimodal-Losses>

2 THEORETICAL BACKGROUND

This chapter provides the necessary theoretical background to understand multimodal learning from a technical perspective. It starts with a presentation of the Transformer architecture which is the core of most modern multimodal models. The section describes how the Transformer was transferred from the language to the vision domain and ultimately applied in multimodal contexts. It discusses why and how Transformers are so effective in this domain, expanding on the architectural differences in single-stream versus dual-stream designs. It then presents the loss functions that are commonly found in the multimodal literature and illustrates these components with specific examples of multimodal models, with a special emphasis on CLIP. The chapter ends with a discussion of related work.

2.1 TRANSFORMER ARCHITECTURE

Since its introduction in 2017, the Transformer architecture has become one of the most widely used models in deep learning. [VSP⁺, XZC] Originally developed for machine translation, it soon became the dominant architecture in Natural Language Processing (NLP), largely due to its role in enabling the development of large language models (LLMs). These models have demonstrated strong generalization across a wide range of NLP tasks, including question answering, summarization, and text generation. [DCLT, BMR⁺20] The influence of the Transformer has also extended beyond NLP, with successful adaptations to other domains such as computer vision. [DBK⁺]

At its core, the Transformer is a sequence-to-sequence model that differs fundamentally from earlier approaches like recurrent neural networks (RNNs) or LSTMs. Instead of modeling sequences through recurrence, the Transformer uses a self-attention mechanism to directly capture dependencies between all elements of the input sequence in parallel. Rather than processing the input step-by-step, every token in a sequence can attend to every other token simultaneously. As a result, the model can learn long-range dependencies more effectively and is more parallelizable on modern hardware, despite the quadratic complexity of the self-attention mechanism with respect to input length.

2 Theoretical background

The basic building block of the Transformer is the Transformer layer, which consists of two main sub-layers: a multi-head self-attention layer, and a point-wise feedforward layer. Each sub-layer is wrapped in a residual connection and followed by layer normalization. In the feedforward layer, the same two-layer MLP is applied independently to each token’s representation.

The self-attention mechanism computes a weighted average of all tokens in the sequence for each token, where the weights are determined by a learned compatibility function between the token and every other token.

Concretely, for a given sequence of token embeddings $X \in \mathbb{R}^{n \times d}$, self-attention is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

where $Q = XW_Q$, $K = XW_K$, and $V = XW_V$ are the query, key, and value projections of the input X , and d_k is the dimension of the key vectors. The softmax ensures that attention weights are normalized. Multi-head attention replicates this mechanism multiple times in parallel with different learned projections, allowing the model to jointly attend to information from different representation subspaces.

Since the attention mechanism itself is permutation-invariant, position embeddings are added to the input token embeddings to inject information about token order.

The original Transformer architecture is composed of an encoder and a decoder. The encoder maps an input sequence to a sequence of hidden representations. The decoder takes this representation and generates an output sequence autoregressively. The decoder contains an additional self-attention layer that is masked to prevent attending to future tokens, as well as a cross-attention layer that attends to the encoder’s outputs. Since only the encoder is relevant for the models used in this thesis, a detailed description of the decoder side is omitted. Readers are referred to the original paper for a full exposition. [VSP⁺]

2.1.1 BERT

A slightly modified version of the original Transformer architecture was introduced in 2018 under the name BERT, short for Bidirectional Encoder Representations from Transformers. [DCLT] Unlike the original Transformer, which includes both an encoder and a decoder and was developed for sequence-to-sequence tasks like machine translation, BERT uses only the encoder stack. This enables it to learn deep bidirectional representations by attending to both left and right context in a text sequence. BERT is pretrained in a self-supervised manner using two objectives: Masked Language Modelling (MLM), where random tokens are masked and

predicted, and Next Sentence Prediction (NSP), where the model predicts whether two input sentences follow each other. In contrast to the original Transformer, which was trained end-to-end for a specific task, BERT was designed as a general-purpose pretraining framework. Its learned representations can be fine-tuned with minimal task-specific changes for a wide range of downstream tasks such as classification, question answering, and named entity recognition. The dual-stream multimodal models used in this thesis adopt a BERT-like architecture as the text encoder, which closely follows the standard Transformer encoder design.

Thanks to this huge advancement in the NLP domain, machine learning researchers also looked for ways to apply it to other domains, one of them being Computer Vision (CV).

2.1.2 VISION TRANSFORMER

Following the success of the Transformer in the NLP domain, the machine learning community began exploring how to extend the architecture to other modalities, particularly vision. Early approaches often focused on integrating attention mechanisms into convolutional neural networks or combining both architectures in hybrid models.[XZC] In contrast, the Vision Transformer (ViT) [DBK⁺] introduced a simpler design by applying the standard Transformer encoder directly to image data with minimal architectural changes. The main difference compared to language models lies in the input processing: instead of tokenizing text, ViT splits an image into fixed-size patches, flattens them, and linearly projects each patch into an embedding vector to form a token sequence. Figure 2.1 illustrates the design.

The Vision Transformer is, for the most part, a standard Transformer encoder that processes a sequence of image patches through multiple layers of multi-head self-attention and feedforward networks. Input images are divided into non-overlapping, fixed-size patches, which are flattened and linearly projected into vectors matching the model’s embedding dimension. These patch embeddings are then summed with learnable positional embeddings to retain information about their position in the sequence. The positional embeddings are one-dimensional and correspond to the linear order of the patch sequence, rather than the original two-dimensional grid structure of the image.

Slight architectural modifications to the original Transformer encoder include applying layer normalization before, rather than after, each sub-layer, and using a GELU activation instead of ReLU in the feedforward layers. Additionally, some implementations omit dropout or reduce it significantly, and may adjust initialization schemes or use learnable scaling factors to improve stability during training

Although Vision Transformers lack the inductive biases inherent to convolutional neural networks, such as locality and translation equivariance, they have been shown to outperform CNNs when trained on large-scale datasets like ImageNet-21k. However, on mid-sized

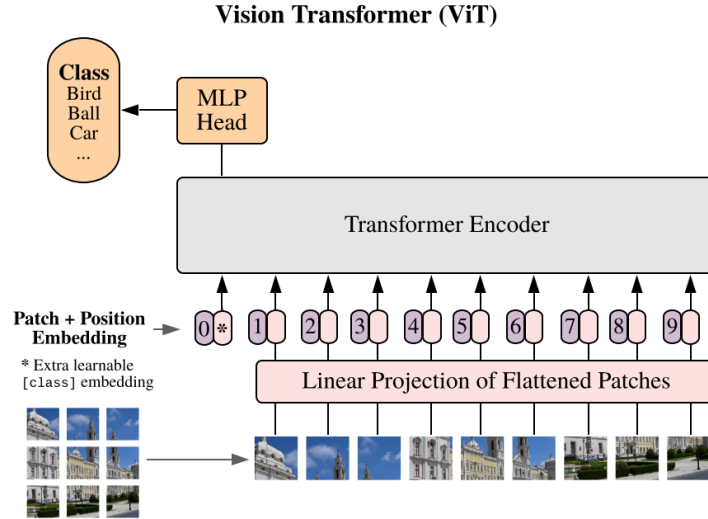


Figure 2.1: **Overview of the Vision Transformer (ViT) architecture.** [DBK⁺] The input image is split into fixed-size patches, each patch is flattened and linearly projected to form patch embeddings, which are then summed with positional embeddings and fed into a standard Transformer encoder.

datasets such as ImageNet-1k, which contains around one million training images, ViTs tend to underperform compared to CNNs. This is because Transformers generally require more training data to learn useful representations, whereas CNNs benefit from their built-in architectural priors. [DBK⁺]

2.2 MULTIMODAL TRANSFORMERS

The successful adaptation of the Transformer architecture from text to vision has shown that it can be applied across different data modalities. A key feature that enables this flexibility is the tokenization and embedding of input data, which allows the architecture to process arbitrary modalities in a unified way. Combined with the self-attention mechanism, which captures dependencies of any length between tokens, this makes the Transformer a powerful backbone for multimodal models. [XZC]

One advantage of Transformer architectures in multimodal settings is their flexible tokenization scheme, which allows different input modalities to be processed in a unified way. [XZC] This process consists of two main steps: tokenization and embedding. Tokenization refers to splitting an input into a sequence of discrete elements. For text, this is typically done using subword-based methods such as WordPiece or Byte-Pair Encoding (BPE) [DCLT], or byte-level tokenization as used in models like GPT [RN18]. For visual inputs,

the two-dimensional structure must first be flattened into a one-dimensional sequence. A common approach, introduced with the Vision Transformer (ViT) [DBK⁺], is to divide the image into fixed-size patches and treat each patch as a visual token. Alternatively, many earlier multimodal models use visual features extracted from pretrained convolutional neural networks, such as region-of-interest (RoI) features from object detectors like Faster R-CNN [RHGS, CZH⁺], which are then linearly projected to form input tokens for the Transformer. The choice of visual tokenization strongly affects the model’s capacity to learn fine-grained visual-linguistic relationships, as explored in works such as ViLT [KSK].

2.2.1 ARCHITECTURE: SINGLE-STREAM VS. DUAL-STREAM

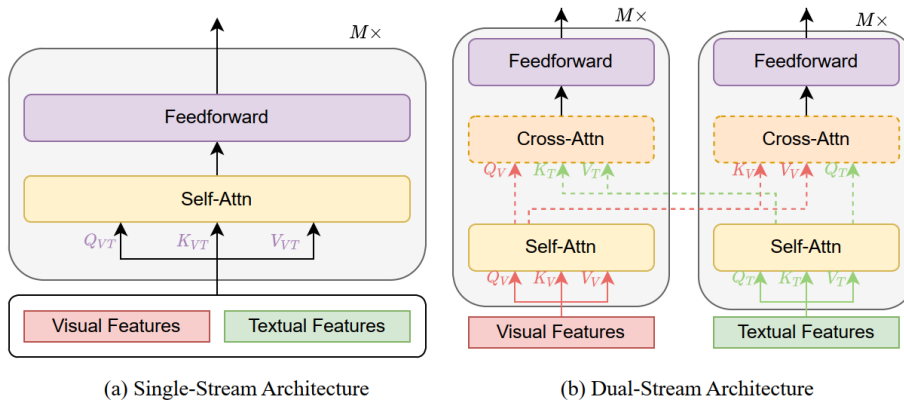


Figure 2.2: Comparison of single-stream and dual-stream multimodal Transformer architectures. [CZH⁺]

Multimodal Transformer models can broadly be categorized into two types: single-stream and dual-stream architectures. [CZH⁺, LZZ⁺, XZC] Single-stream models use a shared Transformer encoder that processes both modalities together within the same layers, enabling early fusion of visual and textual information. In contrast, dual-stream models use separate encoders for each modality, keeping visual and textual representations separate until a later stage. The key difference between these two approaches lies in how cross-modal relationships are computed. In single-stream models, interactions are learned jointly throughout the shared encoder, while dual-stream models rely on a separate fusion mechanism, such as cross-attention, to combine the modalities at a later stage. Figure 2.2 illustrates the differences. In single-stream models, both modality inputs are processed jointly by a shared Transformer encoder, and the relationship between the modalities is learned through the self-attention layers of the standard architecture. A key design choice is how the modalities are fed into the encoder. One option is to combine the token embeddings from each modality through

position-wise summation. The other is to concatenate the token sequences and treat them as a single input stream. While concatenation is more computationally expensive due to the quadratic complexity of self-attention, it enables a more fine-grained modeling of inter-modal dependencies. In the dual-stream setting, the standard self-attention mechanism is extended to cross-attention to enable interaction between modalities. The key difference is that in self-attention, the queries, keys, and values all come from the same input sequence, while in cross-attention, the queries are derived from one modality and the keys and values come from the other. This allows one modality to attend to relevant information in the other, enabling targeted information exchange between the separate encoders. Rather than choosing strictly between a single-stream or dual-stream architecture, some multimodal models combine both approaches. [XZC] One common design is to stack a single-stream encoder on top of a dual-stream model, where the outputs of the separate modality encoders are concatenated and then jointly processed by shared Transformer layers. The reverse is also possible: the concatenated inputs are first passed through a single-stream encoder, and the resulting joint representation is then split and processed further by modality-specific encoders. These hybrid designs aim to balance the benefits of early and late fusion.

2.2.2 LOSS FUNCTIONS IN MULTIMODAL LEARNING

Since the introduction of the Transformer architecture and its success in NLP pre-training, researchers have tried to transfer this approach to vision and vision-language learning. [XZC, CZH⁺] Many multimodal architectures have been proposed since then, which can be grouped into single-stream and dual-stream models. These models are trained with different objectives, often referred to as pretext tasks. Pretext tasks are not the final tasks the model is designed to solve, but are used to help the model learn useful representations that can later be applied to downstream tasks like visual question answering, visual reasoning, image classification, image captioning, image-text retrieval, visual entailment, or referring expression grounding. [CZH⁺]

After categorizing a multimodal model as either single-stream or dual-stream, the next main difference between models of the same type lies in the choice of pretext tasks or loss functions used during training. These loss functions can be broadly grouped into categories such as completion, matching, and temporal losses. [CZH⁺] The temporal category mainly applies to video data. In addition to these groups, there are also specific loss functions that do not fit neatly into any of them.

Completion losses are also referred to as masking losses, as their objective is to predict tokens or features that have been masked at the input, based on the context of the remaining sequence elements. [CZH⁺] In the text domain, this task is known as Masked Language Mod-

elling. In the visual domain, similar objectives include Masked Image Region Regression, where the goal is to reconstruct the original feature vector of a masked region, and Masked Image Region Classification, where the model predicts the object category of the masked region. These masking objectives are commonly used to learn strong unimodal representations. In a multimodal setting, they can be extended by conditioning the prediction on the other modality, resulting in tasks like Image-Conditioned Masked Language Modelling or Text-Conditioned Masked Region Classification.

Matching losses aim to teach the model which multimodal pairs in a batch belong together and which do not. [CZH⁺] A commonly used loss in this category is the Image-Text Matching (ITM) loss. The objective in ITM is to predict whether a given image-text pair is a match. Negative pairs are created by replacing either the image or the text in a matching pair with a sample from another pair in the batch. In single-stream models, the prediction is usually based on the [CLS] token of the joint input sequence. In dual-stream models, the final embeddings of both modalities are combined, for example by concatenation or another fusion method, and passed to a classification head for prediction.

Another loss that falls into this category is the contrastive loss. Originally developed for self-supervised learning in unimodal settings, such as in the visual domain with the SimCLR model [CKNH], this type of loss was first applied to vision-language learning in the CLIP model [RKH⁺]. The goal is to embed representations from both modalities into a shared latent space, where matching pairs are placed close together and non-matching pairs are pushed apart. By contrasting positive with negative pairs, the model learns to align semantically similar data across modalities.

Temporal losses form a smaller category of pretext tasks that mainly apply to video-based or sequential data. [CZH⁺] These losses aim to model the temporal structure within a sequence, such as predicting the correct order of frames in a video. While useful for learning temporal dynamics, they are less relevant for most image-text models. Therefore, the majority of loss functions used in multimodal vision-language models fall into the completion or matching categories.

2.3 MODELS

Based on this taxonomy, most Transformer-based multimodal models in the literature can be categorized along two main dimensions: whether they use a single-stream or dual-stream architecture, and which loss functions are used during pretraining. In the following, we present a selection of models to illustrate how these two dimensions are combined in practice.

2.3.1 BERT-INSPIRED MODELS

One of the earliest Transformer-based multimodal models is ViLBERT [LBPL], which stands for “Vision & Language BERT.” ViLBERT is a dual-stream model inspired by the success of BERT and extends the self-supervised pretraining paradigm to the vision-and-language domain. The architecture consists of two separate streams for visual and textual input, which interact through dedicated cross-attention layers that allow each modality to attend to the other. ViLBERT is pretrained with both masking and matching losses. It uses Masked Language Modelling on the text input and a form of masked image region classification, where the model predicts the object class label of a masked region feature extracted from a pretrained object detector. Additionally, it uses an Image-Text Matching (ITM) loss, where the global features of both modalities, represented by their respective [CLS] tokens, are passed to a classification layer to predict whether the image and text pair is a match or not.

Another multimodal model developed around the same time is VisualBERT [LYD⁺], which also builds on the BERT architecture. In contrast to ViLBERT [LBPL], VisualBERT uses a single-stream architecture, encoding both modalities jointly in the same Transformer layers. It is trained with a Masked Language Modelling objective but does not apply any masking to the visual input. In addition, it uses an Image-Text Matching loss, where the [CLS] token of the joint multimodal sequence is used to predict whether the image and text form a matching pair.

UNITER [CLY⁺] is a hybrid-stream model that combines elements of both dual-stream and single-stream architectures. It begins with two separate encoders: a Faster R-CNN [RHGS] for the image input, which extracts region-of-interest (ROI) features, and a BERT-based Transformer for the text input, which produces contextualized word embeddings. The visual features are augmented with positional embeddings to preserve spatial information. After this initial stage, the visual and textual embeddings are concatenated and passed into a shared single-stream Transformer, where cross-modal interactions are learned. UNITER is trained with several objectives: Masked Language Modelling and Masked Region Modelling, both of which are cross-modal due to the single-stream Transformer. It also uses an Image-Text Matching loss. Additionally, it employs a Word-Region Alignment loss, which is a fine-grained matching objective. This loss encourages alignment between individual word embeddings and the most relevant visual region features by maximizing similarity for correct pairs and minimizing it for incorrect ones, usually based on soft or hard alignment labels.

VL-BERT [SZC⁺] is a single-stream vision-language model that feeds both text tokens and visual features, extracted from region-of-interest (RoI) detections using a pretrained Faster R-CNN [RHGS], into a shared Transformer encoder. This design enables early cross-modal fusion, allowing both modalities to interact from the first layer. VL-BERT is pretrained with

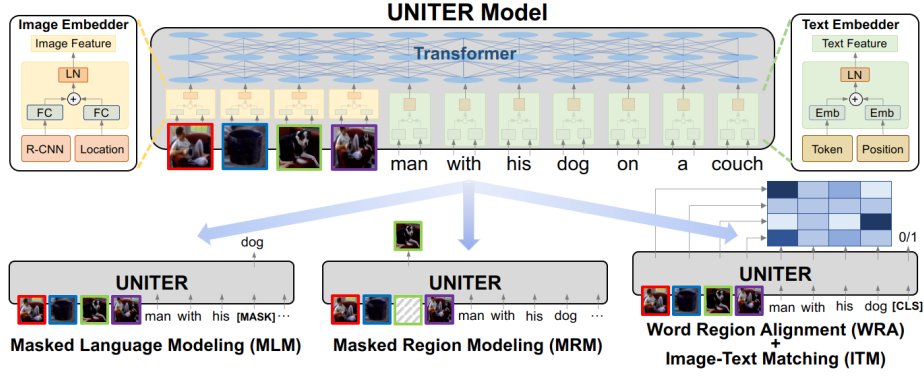


Figure 2.3: **Architecture of UNITER** [CLY⁺] UNITER is a hybrid-stream model that uses two separate streams for each modality before processing with a single-stream Transformer. It is trained with four loss functions: Masked Language Modelling, Masked Region Modelling, Word Region Alignment and Image-Text Matching.

two objectives: Masked Language Modelling (MLM), where text tokens are predicted based on both textual and visual context, and Masked RoI Classification, where the model predicts the object category of a masked visual region using the surrounding text. Pretraining is done on Conceptual Captions [SDGS18] for learning image-text alignment, and on a large text-only corpus with MLM to improve the language representations. Unlike ViLBERT [LBPL], which separates modality streams, VL-BERT learns cross-modal interactions throughout the full depth of the network.

ViLT [KSK] is a single-stream vision-language model that simplifies the multimodal architecture by removing the need for visual feature extractors such as Faster R-CNN [RHGS]. Instead, it applies a linear projection to flattened 2D image patches, following the approach of the Vision Transformer (ViT) [DBK⁺], and combines these with text token embeddings. Both modalities are jointly processed in a single BERT-like Transformer, enabling early and full cross-modal fusion. ViLT is pretrained with three objectives: Image-Text Matching, where the [CLS] token is used to predict whether an image-text pair is aligned; Masked Language Modelling, conditioned on the visual context; and a Word-Patch Alignment loss, which encourages fine-grained associations between individual words and image patches by maximizing similarity between aligned pairs. This setup allows ViLT to achieve strong performance while remaining computationally efficient.

As shown in the previous examples, multimodal models in the literature often rely on extensive and carefully designed pretraining objectives to learn cross-modal representations. These approaches are largely inspired by the success of BERT’s self-supervised pretraining paradigm and aim to adapt it to the vision-language domain. In the following section, we describe a related line of work that follows the same motivation but adopts a much sim-

pler, yet highly effective, approach to learning multimodal representations: the CLIP model. [RKH⁺]

2.3.2 CLIP: CONTRASTIVE LANGUAGE-IMAGE PRE-TRAINING

The CLIP [RKH⁺] model, short for Contrastive Language-Image Pretraining, constitutes an exception to the earlier definition of dual-stream models. The modality interaction is not brought about by cross-attention or fusion of the modality features but by alignment of the modality embeddings in a shared multimodal projection space. This is facilitated by the contrastive loss function which is the defining feature of this model.

Introduced by OpenAI in 2021, CLIP was one of the first large-scale vision-language models to use a contrastive objective for pretraining. Its development was inspired by the success of the GPT family of language models, which were trained in a self-supervised manner on massive web-scale text corpora. [RN18] These models showed remarkable generalization across a wide range of NLP tasks, despite not being explicitly trained on them. While GPT-2 was typically fine-tuned on downstream tasks after pretraining, GPT-3 demonstrated strong zero-shot performance through in-context learning, enabled by both its larger model size and training dataset. [RWC⁺19, BMR⁺20] This success motivated the idea of using a similarly large dataset to train a vision-language model with general-purpose capabilities.

Most vision models prior to CLIP were trained in a supervised manner on labeled image datasets, with ImageNet being the most prominent example. ImageNet [RDS⁺] contains over 14 million human-annotated images and played a central role in advancing computer vision over the past decade. However, scaling up such datasets is expensive and time-consuming. At the same time, the success of the GPT models demonstrated that pretraining on large amounts of uncured text data from the internet can be both feasible and effective. This led to the idea of leveraging similarly large-scale, weakly supervised data sources, specifically images paired with their associated text found online, to train a general-purpose vision-language model.

Another motivation behind this approach is the hypothesis that language supervision from web-scale data can expose the model to a much broader range of visual concepts. Traditional class-labeled datasets like ImageNet divide the visual domain into a fixed set of predefined categories, which limits the diversity of semantic information available during training. In contrast, a large collection of image-text pairs scraped from the web is expected to reflect a wider and more natural distribution of visual concepts, allowing the model to learn more flexible and generalizable representations. [SDT⁺]

The core idea behind CLIP is to use two separate encoders, one for text and one for images, and train them to produce embeddings that are aligned in a shared multimodal space. During

training, the model is given batches of image-text pairs. The goal is to bring the embeddings of matching image and text pairs closer together in the embedding space, while pushing apart the embeddings of non-matching pairs. This is done using a contrastive loss, which encourages high similarity between the correct pairs and low similarity between all other combinations in the batch. In this way, the model learns to associate images and texts that describe the same content, without needing manually defined class labels.

One of the most powerful capabilities enabled by CLIP’s training approach is zero-shot image classification. This refers to the ability to classify images from new datasets without any additional training or fine-tuning. Instead of relying on a fixed classifier trained on specific categories, CLIP constructs a classifier dynamically using natural language. For a given dataset, the class names are converted into textual prompts, typically using a template like “a photo of a class name”, which are then embedded by the text encoder to form a set of class representations. At inference time, an input image is encoded by the image encoder, and its embedding is compared to each of the text embeddings. The class whose text embedding is most similar to the image embedding is selected as the prediction. This allows CLIP to generalize to new tasks and datasets without requiring any labeled examples from the target domain.

Thanks to this design, CLIP demonstrated strong performance across a wide range of benchmarks, including image classification, image-text retrieval, and robustness evaluations. [RKH⁺] It showed competitive or even state-of-the-art results on many zero-shot classification tasks, and it was among the first models to exhibit strong performance without fine-tuning across dozens of datasets. Additionally, CLIP was shown to be more robust to distribution shifts and adversarial examples compared to traditional supervised models. These results highlighted the effectiveness of contrastive language supervision at scale for training general-purpose vision models.

2.4 RELATED WORK

LOSS ABLATION STUDIES

Other works have studied the effect of loss functions in multimodal models, particularly in dual-stream CLIP-like architectures like the one used in this work.

SLIP [MKWX] extends the CLIP architecture by adding image-side self-supervision alongside the image-text contrastive loss. Like in this work, SLIP adds the SimCLR [CKNH] loss which encourages the model to learn transformation-invariant visual features through an image-side contrastive loss. The model uses separate projection heads: one for computing the image embeddings used in the CLIP contrastive loss, and another for the SimCLR

loss. The authors conduct an ablation study to assess the improvement by using additional SimCLR. SLIP demonstrates improved performance on several benchmarks, including zero-shot classification on ImageNet and linear probing accuracy. In contrast, the approach taken in this thesis applies the SimCLR loss directly at the same projection head used for the CLIP loss. This means both objectives act on a shared embedding space.

Another work that exhibits parallels to this thesis is DeCLIP [LLZ⁺], short for *Data-efficient CLIP*. DeCLIP extends the original CLIP framework by incorporating additional self-supervision objectives for each modality: Masked Language Modelling (MLM) on the text side and view-based contrastive learning on the image side. It also introduces cross-modal multi-view supervision by applying stochastic augmentations to text captions, enabling the model to contrast multiple views of each modality. This effectively doubles the amount of cross-modal signal available during training. In addition, DeCLIP uses nearest-neighbor supervision on the text side, where semantically related captions are retrieved using nearest-neighbor search in the embedding space and used to form additional positive pairs. The authors show that this combination of objectives allows DeCLIP to achieve performance comparable to CLIP while using roughly seven times less training data, and to outperform CLIP when scaled further.

An ablation study in the paper examines the impact of the different types of supervision. Unlike the approach in this thesis, the authors do not ablate the unimodal self-supervision losses individually, but only evaluate their combined effect relative to using the contrastive image-text loss alone. Another difference lies in the choice of the image-side loss: DeCLIP employs SimSiam for self-supervised learning, a method that, unlike contrastive approaches such as SimCLR, does not require negative samples. [CH20]

RESEARCH ON LANGUAGE SUPERVISION IN MULTIMODAL MODELS

"Does language help generalization in vision models?" [DCBV] is a systematic study between the capabilities of multimodal models like CLIP and unimodal vision models. The aim of the study was to investigate whether the language supervision provided by training on image-caption datasets conferred better generalization than training only on class-labelled image datasets, since the idea is that language provides more semantic information than image datasets alone and can thus help with semantic grounding of the visual feature space. Through extensive tests in few-shot learning, transfer learning and unsupervised clustering, the authors found that multimodal models don't generalize better than vision-only models. They also conducted model similarity analysis using Representational Similarity Analysis (RSA) [KMB08], showing that representations learned by multimodal models are different from vision-only models. A limitation of the study is that they compare pre-trained models of different sizes

and architectures, which were pre-trained on different datasets. This thesis also investigates the influence of language supervision versus image-only supervision by comparing downstream task performance between models trained only with SimCLR and models trained with CLIP on the same underlying dataset, while controlling for all other training variables. We show that training with language-supervision does confer performance benefits, as measured by linear probing.

"Is a caption worth a thousand images?" [SDT⁺] is a more controlled investigation of the role of language supervision in learning visual representations. The authors study the effect of training a model with CLIP vs SimCLR. The key insight is that both loss functions are based on contrasting positive pairs with negative pairs. Pairs of examples are constructed by generating a different view of an image. In CLIP, the view is the caption, while in SimCLR the view is an augmented version of the image. The authors control for all other variables between the models. Their key finding is that CLIP indeed learns better representations than SimCLR, as measured by performance on linear probing task, when trained with more data. This thesis replicates that finding, and we additionally show that training with CLIP and SimCLR can have synergistic effects that outperform both approaches in a low-data regime.

3

EXPERIMENTAL DESIGN

This chapter outlines the details of the experimental design employed in this work.

Section x details the dual-stream architecture we used for this experiment. Section x gives a description of the pre-training dataset. The theoretical formulation as well as the concrete implementation for every evaluated loss functions is given in Section x. Section x explains the theory on the evaluation methods we used, namely the downstream tasks and Centered Kernel Alignment (CKA). The chapter concludes in Section x with details on the model training and evaluation setup.

3.1 MODEL ARCHITECTURE

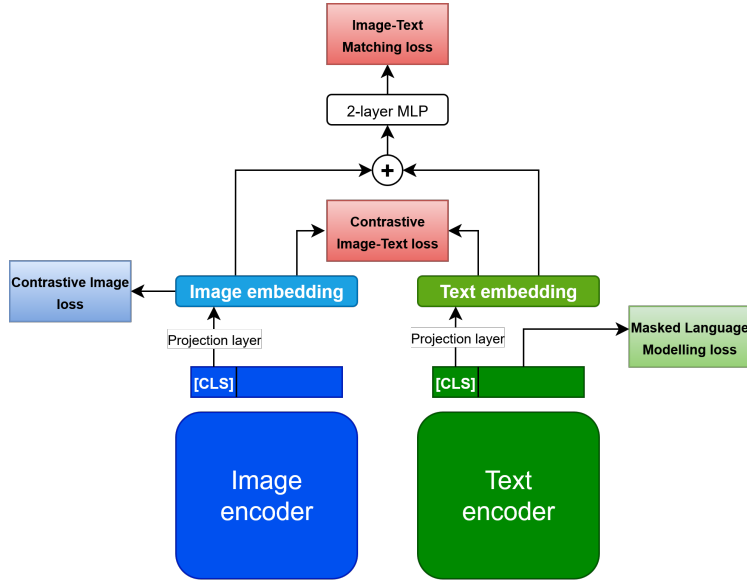


Figure 3.1: **Diagram of the dual-stream model architecture.** Images and texts are processed independently by their respective encoders. [CLS] tokens of the last encoder layers are projected into a shared embedding space, where the Contrastive Image-Text loss is applied. A concatenation of the embeddings is processed with 2-layer MLP before applying the Image-Text Matching loss. Masked Language Modelling loss is applied over the whole text token sequence. SimCLR loss is applied at the image embedding layer.

3 Experimental design

The model architecture used in this study is a simple dual-stream encoder design. Figure 3.1 shows a diagram of the architecture.

As the goal of this work is to study the effect of loss functions through a controlled ablation study, the choice of architecture depends on the loss functions that are chosen for study. Since one of the loss functions investigated in this work is the CLIP loss, we have to choose an architecture that intentionally avoids the use of cross-attention layers or fusion modules. CLIP necessitates that the two modality-specific streams, image and text, operate independently throughout their respective encoders, and that interaction between the streams happens only at the final projection stage, where the pooled outputs of the encoders are mapped into a shared embedding space using a linear projection head. This is a limitation of this study that is addressed in the conclusion of this work.

Most importantly, we deliberately initialize both encoders from scratch in all our experiments. This choice ensures a controlled experimental setup, allowing us to properly isolate and assess the influence of the individual loss functions on representation learning and downstream task performance.

The architecture is implemented using the `VisionTextDualEncoderModel` class from the HuggingFace Transformers library. This model class provides a flexible interface for combining different Transformer-based models from the HuggingFace model hub, either initialized from scratch or using pre-trained weights.

Each encoder is equipped with a single linear layer followed by a $\tanh$ nonlinearity (pooling layer) that extracts the hidden state corresponding to the [CLS] token from the last encoder layer. The resulting pooled output is passed through a linear projection layer to obtain the final embedding for the shared multimodal space.

For the visual encoder, we use the architecture of `google/vit-base-patch16-224`, a Vision Transformer Base model with 12 layers, hidden size of 768, and 12 attention heads. It processes images at 224×224 resolution using 16×16 non-overlapping patches.

The text encoder follows the `google-bert/bert-base-uncased` architecture, which consists of 12 layers, hidden size of 768, and 12 attention heads, trained on uncased English text.

We make use of the pre-trained processor classes provided by HuggingFace for both encoders to ensure consistent and standardized preprocessing.

For the image encoder, we use the `ViTImageProcessor` with standard settings: images are resized to 224×224 pixels and normalized using the default mean and standard deviation values, which are 0.5 across all input channels.

For the text encoder, we use the `BertTokenizer` with the default BERT vocabulary. It utilizes WordPiece tokenization and lowercases all input text.

Table 3.1: Architectural details of the components used in the dual-stream model

Component	Modality	# Parameters	# Layers	Hidden Size	Attention Heads
ViT-Base	Image	86M	12	768	12
BERT-Base	Text	110M	12	768	12
Projection Layer	Multimodal/Image	1.2M	1	512	–
ITM Head (2-layer MLP)	Multimodal	0.5M	2	512	–
MLM Head	Text	30M	1	768	–
Total	–	227.7M	–	–	–

3.2 LOSS FUNCTIONS

3.2.1 MULTIMODAL LOSS: IMAGE-TEXT CONTRASTIVE (CLIP)

The Image-Text Contrastive loss function (which we abbreviate as CLIP) is a multi-modal contrastive loss that trains an image encoder and a text encoder to embed representations of images and corresponding texts in a shared vector space. The core idea behind the loss is to align embeddings of actual image-text pairs closely together in this shared space while simultaneously pushing embeddings of non-matching pairs further apart.

To achieve this alignment, CLS token embeddings, representing the entire image or text from the last layer of the respective encoder, are first projected into a shared, lower-dimensional space. These projected embeddings are then normalized to unit length:

$$\mathbf{v}_i = \frac{\text{Proj}(\text{ImageEnc}(I_i))}{|\text{Proj}(\text{ImageEnc}(I_i))|}, \quad \mathbf{t}_i = \frac{\text{Proj}(\text{TextEnc}(T_i))}{|\text{Proj}(\text{TextEnc}(T_i))|} \quad (3.1)$$

where I_i and T_i represent the i^{th} image and text pair in a batch, respectively. Normalization ensures the dot product corresponds directly to cosine similarity.

The cosine similarity between each pair of image and text embeddings within a batch is computed via matrix multiplication, resulting in an $N \times N$ similarity matrix $\mathbf{L}$:

$$\mathbf{L} = \frac{\mathbf{V}\mathbf{T}^\top}{\tau}, \quad (3.2)$$

where $\mathbf{V} \in \mathbb{R}^{N \times d}$ is the batch of normalized image embeddings, and $\tau > 0$ is a learnable temperature parameter scaling the similarities.

3 Experimental design

Each entry $\mathbf{L}_{ij}$ represents the scaled cosine similarity between the i^{th} image embedding and the j^{th} text embedding.

The loss consists of two symmetric cross-entropy terms:

Image-to-Text Loss ($\mathbf{I} \rightarrow \mathbf{T}$):

$$\mathcal{L}_{\mathbf{I} \rightarrow \mathbf{T}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\mathbf{L}_{ii})}{\sum_{j=1}^N \exp(\mathbf{L}_{ij})} \quad (3.3)$$

Text-to-Image Loss ($\mathbf{T} \rightarrow \mathbf{I}$):

$$\mathcal{L}_{\mathbf{T} \rightarrow \mathbf{I}} = -\frac{1}{N} \sum_{j=1}^N \log \frac{\exp(\mathbf{L}_{jj})}{\sum_{i=1}^N \exp(\mathbf{L}_{ij})} \quad (3.4)$$

The final loss is the average of these two terms:

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{2} (\mathcal{L}_{\mathbf{I} \rightarrow \mathbf{T}} + \mathcal{L}_{\mathbf{T} \rightarrow \mathbf{I}}) \quad (3.5)$$

This formulation encourages embeddings of matching image-text pairs (diagonal entries in $\mathbf{L}$) to have high similarity scores, thus pulling these pairs together. In contrast, non-matching pairs (off-diagonal entries) receive lower similarity scores, effectively pushing them apart in the embedding space.

By explicitly contrasting matched versus unmatched pairs, this loss function enables learning of robust and semantically meaningful embeddings, enabling zero-shot capabilities and effective downstream transfer to different tasks and datasets.

3.2.2 MULTIMODAL LOSS: IMAGE-TEXT MATCHING

The second loss function we consider for our experiments is the image-text matching loss. Image-text matching is a common objective in vision-language learning. It encourages a model to distinguish between matching and non-matching image-text pairs, by learning to assign a higher matching score to aligned pairs. Matching pairs are the actual example pairs from the training dataset, and non-matching pairs are constructed by sampling either a different image or text from the batch.

The implementation of this objective depends on the underlying model architecture. In single-stream models, modalities are fused early by concatenating the image and text tokens into a single sequence. The model captures the relationship between modalities through self-attention across the whole sequence. A common approach in this case is to prepend a special

token to the sequence that learns a global representation of the image-text pair, which is then used as input to the matching head.

In dual-stream architectures, the modalities are processed separately at first, each with their own encoder, and only fused later in the network. This late fusion is typically achieved with cross-attention or co-attentional layers that enable interaction between the modality-specific representations. The matching score is then predicted based on the fused representation after these cross-modal interactions.

The implementation of the ITM loss in this study differs from the typical approach found in dual-stream architectures. Our chosen architecture does not include any cross-attention layers, and cross-modal interaction is restricted to the final layer of the network, where image and text embeddings are aligned by maximizing their cosine similarity. Instead of using an explicit fusion module, we concatenate the normalized image and text embeddings from the projection layer and pass this combined representation to a separate matching head.

IMPLEMENTATION

The matching head is a two-layer multilayer perceptron that receives the concatenated image and text embeddings of the final projection layer as input. Let $\mathbf{v} \in \mathbb{R}^D$ and $\mathbf{t} \in \mathbb{R}^D$ denote the normalized image and text embeddings from the projection layer. The concatenated input is given by $\mathbf{z} = [\mathbf{v}; \mathbf{t}] \in \mathbb{R}^{2D}$. The MLP then computes the matching score as follows:

$$\begin{aligned}\mathbf{h} &= \text{ReLU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) \\ \mathbf{s} &= \mathbf{W}_2 \mathbf{h} + \mathbf{b}_2\end{aligned}$$

where $\mathbf{W}_1 \in \mathbb{R}^{D \times 2D}$, $\mathbf{b}_1 \in \mathbb{R}^D$, $\mathbf{W}_2 \in \mathbb{R}^{2 \times D}$, and $\mathbf{b}_2 \in \mathbb{R}^2$. The output $\mathbf{s} \in \mathbb{R}^2$ represents the unnormalized logits for the binary classification task of predicting whether the image-text pair is matching or not matching.

To construct negative examples, we replace the texts of 50% of image-text pairs in a mini-batch with hard-negatives.

HARD-NEGATIVE MINING

Inspired by ALBEF [LSG⁺], we use hard-negative mining to construct the non-matching pairs for the ITM loss. Since using all negative examples that are available in a batch for training the ITM head would have a quadratic cost, we are forced to limit our choice of negative pairs to train with. To mitigate this, we can rely on the learned similarity at the embedding layer to sample hard negatives.

3 Experimental design

A negative pair is considered hard when the image and text do not form an exact match, but still share some semantic similarity. These types of negatives are more informative during training, as the model needs to learn finer distinctions between similar but incorrect pairs.

We construct such hard-negative examples by exploiting the shared multimodal embedding space at the final layer of the model. Since the image and text embeddings at this layer are aligned through the contrastive loss, their cosine similarity reflects semantic closeness. For each image in a batch, we compute the similarity scores to all text embeddings using the dot product between the normalized embeddings. Applying a softmax over these similarity scores gives a probability distribution from which we can sample a text to construct a hard-negative pair.

Let $\mathbf{v}_i \in \mathbb{R}^D$ denote the normalized image embedding for image i , and let $\mathbf{t}_j \in \mathbb{R}^D$ denote the normalized text embeddings for all texts j in the batch. We compute the similarity scores using the dot product:

$$s_{ij} = \mathbf{v}_i^\top \mathbf{t}_j$$

To obtain a probability distribution over negative candidates, we apply the softmax function:

$$p_{ij} = \frac{\exp(s_{ij})}{\sum_{k=1}^N \exp(s_{ik})}$$

A hard-negative text $\tilde{t}_i$ for image i is then sampled from this distribution:

$$\tilde{t}_i \sim p_{ij}$$

This way, non-matching texts that are more similar to the image have a higher chance of being selected, making the task of distinguishing matching from non-matching pairs more challenging and informative for the model.

3.2.3 UNIMODAL IMAGE LOSS: SIMCLR

The first unimodal loss function we consider is SimCLR. [CKNH] SimCLR stands for "*A Simple Framework for Contrastive Learning of Visual Representations*". Like the image-text contrastive loss discussed earlier, SimCLR is a contrastive loss function. Contrastive losses are based on the idea of pulling positive (matching) example pairs closer together in the representation space, while pushing negative (non-matching) pairs further apart. This leads to a representation space where similar examples are close together and dissimilar examples are far apart. While the CLIP loss contrasts image-text pairs, SimCLR operates only on images.

The key idea is to generate different views of the same underlying image by applying stochastic data augmentations. To do this, an augmentation module applies random transformations to the image, sampling different parameters each time. These transformations come from the same augmentation family, but are applied independently, resulting in two different, yet related, views of the same image. These two augmented views form a positive pair in the contrastive loss. Negative pairs are formed between each view of an image and the two views of all other images in the batch. This means that for every positive pair, there are $2 \times (N - 1)$ negative pairs in a batch of size N , which significantly increases the number of contrastive comparisons. The effect of SimCLR is that the model learns representations that are invariant to the applied transformations. It encourages the model to represent images with similar underlying content in a consistent way, even when those images are perturbed or altered by augmentations.

Formally, let $\mathbf{z}_i$ and $\mathbf{z}_j$ be the normalized representations of two augmented views of the same image. The contrastive loss for a positive pair (i, j) is defined as:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_k \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)} \quad (3.6)$$

where $\text{sim}(\mathbf{z}_i, \mathbf{z}_j) = \mathbf{z}_i^\top \mathbf{z}_j$ is the cosine similarity between the two normalized vectors, and τ is a temperature parameter.

The final loss is computed by summing over all N positive pairs in the batch, where each image has two views and therefore contributes to two symmetric loss terms:

$$\mathcal{L} = \frac{1}{2N} \sum_{k=1}^N [\ell_{2k-1,2k} + \ell_{2k,2k-1}] \quad (3.7)$$

This formulation ensures that the loss accounts for both directions of the positive pairs and provides a consistent signal for learning invariant representations.

In practice, a projection head $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is used to map the representations $b \in \mathbb{R}^d$ learned by the base encoder to a separate embedding space $z = f(b) \in \mathbb{R}^k$, where the loss is applied.

IMPLEMENTATION

In this work, the SimCLR loss is applied to the final projection layer of the dual-stream model, where the CLIP loss is also applied, i.e., $\mathcal{L}_{\text{SimCLR}} = \mathcal{L}_{\text{CLIP}}$.

By doing this, we hope to see an improvement in the quality of image representations in the multimodal embedding space, as this layer is also encouraged to learn transformation-invariant image features. But as we observe in practice, this choice leads to a decrease in rep-

3 Experimental design

resentation quality and downstream task performance, suggesting that the two objectives interfere with each other during optimization. This is discussed in more detail in the discussion section 4.6.

AUGMENTATION MODULE We use the RandAugment [CZSL] module as our main image augmentation method, both for general data augmentation during pre-training and for generating augmented views in the SimCLR loss. RandAugment works by randomly selecting a fixed number of image transformations from a predefined list and applying them to each image. For every image, a new combination of augmentations is sampled. Two parameters control its behavior: the number of transformations applied and the strength with which they are applied.

The set of possible augmentations includes operations such as AutoContrast, Equalize, Rotate, Solarize, Color, Posterize, Contrast, Brightness, Sharpness, ShearX/Y, and TranslateX/Y. The transformations are applied in sequence and change the appearance of the image while keeping its semantic content intact.

We use parameters `num_ops=7` and `magnitude=9` for our main models, and `num_ops=3` with `magnitude=8` for our small models in the dataset size ablation study.

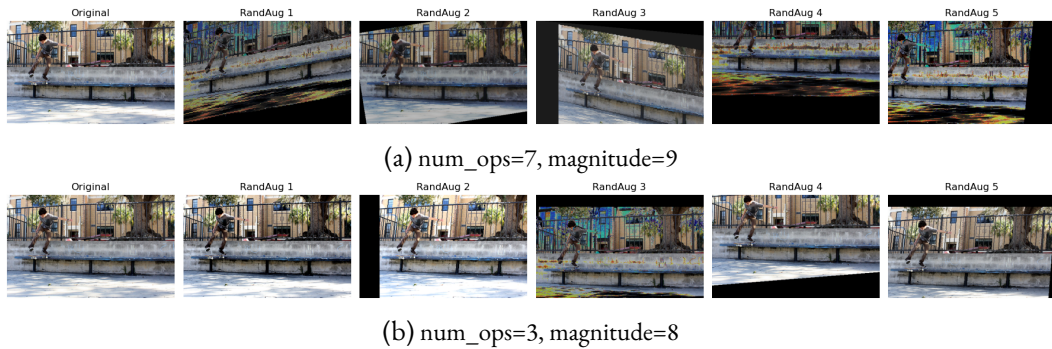


Figure 3.2: Examples of RandAugment image augmentation for two different settings. Leftmost image is original, rest are augmented versions.

3.2.4 UNIMODAL TEXT LOSS: MASKED LANGUAGE MODELLING

The last loss function we consider is the Masked Language Modelling (MLM) loss, which was popularized by the BERT [DCLT] model. The objective of this loss is to predict the correct token from the vocabulary at positions that are masked in the input sequence. To create the masked input, a proportion of tokens in the sequence is randomly selected. These selected

tokens are either (1) replaced by a special [MASK] token, (2) substituted with a random token from the vocabulary, or (3) left unchanged.

After passing the masked input through the encoder, the resulting token representations are projected through a linear classification head to predict the original tokens. The loss is computed as the cross-entropy between the predicted token distribution and the true token at the masked position:

$$\mathcal{L}_{\text{MLM}} = - \sum_{i \in \mathcal{M}} \log p(t_i | \hat{h}_i) \quad (3.8)$$

Here, $\mathcal{M}$ denotes the set of masked token indices, t_i is the original token at position i , and $\hat{h}_i$ is the contextualized embedding at the masked position after the final encoder layer. The linear classifier maps $\hat{h}_i$ to a distribution over the vocabulary from which the prediction is made.

By utilizing this objective, the text encoder learns to encode tokens based on the bidirectional context in a sequence. This is different from the autoregressive language modelling loss, where the objective is to predict the next token in a sequence based only on the context of the preceding tokens. Unlike the latter loss function, MLM does not enable generative tasks like producing new text from a given sequence.

Instead, MLM enables models to develop a deeper understanding of language structure and semantics. It is particularly effective for learning rich contextual representations that are useful for a wide range of downstream tasks such as classification, question answering, and textual entailment.

This loss, when combined with the next sentence prediction (NSP) objective, formed the pre-training strategy of BERT. [DCLT] This combination proved to be highly effective and contributed significantly to BERT’s success as a general-purpose model for a wide range of downstream NLP tasks.

IMPLEMENTATION

We apply the single-linear layer MLM head to the last hidden sequence of the text encoder.

We select 20% of the input tokens for masking. Out of the selected tokens, 80% are replaced by the [MASK] token, 10% are replaced by a random token from the vocabulary, and the remaining 10% are left unchanged. This follows the masking strategy outlined in the original BERT paper [DCLT] and has become a standard approach for applying the MLM objective.

Different from other multimodal architectures such as UNITER [CLY⁺] or VL-BERT [SZC⁺], our implementation of the MLM objective is not conditioned on input from the other modality, i.e., images. These models apply the masking loss to a joint multimodal repre-

sensation, enabled by cross-attention layers, that allow the text encoder to incorporate visual context. In contrast, since we do not employ cross-attention layers in our model, the MLM loss remains entirely unimodal and is applied independently to the text encoder without any interaction with the visual input. This is a limitation that is brought about by our chosen architecture, as we chose a CLIP-like dual-stream encoder architecture to accommodate the CLIP loss.

3.3 TRAINING DATA

The pre-training dataset used in this study is a concatenation of three publicly available image-caption datasets: Conceptual Captions 3M, MS COCO Captions, and Visual Genome. Each of these datasets contains natural images paired with descriptive text, and together they provide a diverse and relatively large-scale source of supervision for training multimodal models.

Conceptual Captions 3M (CC3M) [SDGS18] is the largest of the three datasets, comprising approximately 3.3 million image-text pairs. It is automatically collected from the web by extracting image-description pairs from web pages, followed by heuristic filtering and text normalization. Due to its size and diversity, CC3M contributes the majority of the training data in this work.

MS COCO Captions [CFL⁺] contains roughly 118,000 images, each annotated with five human-written captions. This results in about 590,000 image-caption pairs. The captions are short, descriptive, and focused on everyday scenes, offering high-quality alignment between visual and textual modalities. During training, we sample one of the five captions for each image at random.

Visual Genome [KZG⁺] provides dense region-level captions for a collection of natural images, originally comprising around 108,000 images. For this study, we use the LLaVA [LLWL] split, which filters out images that overlap with CC3M and MS COCO to avoid duplicate training examples. After filtering, this results in approximately 60,000 unique image-caption pairs from Visual Genome.

Combining all three sources yields a pre-training dataset of roughly 3.95 million image-caption pairs. Conceptual Captions accounts for the largest portion of the data, followed by MS COCO and then Visual Genome.

Dataset	Image-Caption Pairs
Conceptual Captions 3M (CC3M)	3,300,000
MS COCO Captions	590,000
Visual Genome (LLaVA split)	60,000
Total	3,950,000

Table 3.2: Number of image-caption pairs per dataset used for pre-training.

3.4 EVALUATION METHODS

3.4.1 ZERO-SHOT IMAGE CLASSIFICATION

The first downstream task we evaluate the models on is zero-shot image classification. Zero-shot image classification refers to the task of classifying images from a dataset that was not seen during pre-training, without any training or fine-tuning on that dataset. The term “zero-shot” describes this property: the model is expected to correctly classify examples from classes it has never seen before, not even a single image.

This ability is made possible by the language understanding capabilities of the text encoder, and the alignment of text and image representations in the shared embedding space. During training, the model learns to map both modalities into the same vector space at the final projection layer, so that matching image-text pairs are close to each other.

To do zero-shot classification, a classifier must be constructed in the embedding space. This is done by turning the class labels of the new dataset into textual descriptions, for example using templates like “a photo of a class”. Each class name is embedded by the text encoder, and the resulting embeddings are normalized. These serve as the classifier weights. Each image in the dataset is then encoded using the image encoder, and the resulting embedding is also normalized. The prediction is done by computing the dot product between the image embedding and every class embedding. Because of the normalization step, the dot product is equal to the cosine similarity, and the class with the highest similarity is selected.

This procedure mirrors the training setup of the contrastive image-text loss, but is now used at inference time. Zero-shot classification directly exploits the learned multimodal embedding space, where the model has been trained to place aligned image-text pairs close together and unrelated pairs further apart. In contrast to fine-tuning or linear probing, there is no adaptation to the specific dataset used for testing. This makes zero-shot classification a strong indicator of how generally useful and transferable the learned representations are out of the box.

3 Experimental design

However, this also makes the task much more dependent on the distribution of the pre-training dataset. If certain concepts or domains were not well represented during pre-training, it will be difficult for the model to recognize them. Fine-grained datasets, or datasets from a very specific domain, will be more challenging to classify than with approaches that use some form of adaptation like fine-tuning or few-shot learning. Zero-shot classification is therefore particularly sensitive to coverage and diversity in the pre-training distribution.

CLASSIFICATION WITH IMAGE-TEXT MATCHING HEAD In our experiments, we also evaluate the use of the Image-Text Matching head in zero-shot classification. To do this, we pre-select k predictions via contrastive similarity and select the final prediction with the matching head for models that have been trained with ITM.

3.4.2 ZERO-SHOT IMAGE-TEXT RETRIEVAL

The second task we evaluate the models on is zero-shot image-text retrieval. Similar to zero-shot image classification, zero-shot image-text retrieval leverages the shared multimodal embedding space at the model’s last layer to match images and text from a test dataset. Image-text retrieval refers to the task of matching an image with a sentence that describes its content. The task can be performed in both directions: in image retrieval, the model searches for an image that best matches a given description; in text retrieval, the model searches for a description that best matches a given image. This differs from image captioning, as the model does not generate the description, but instead matches an existing one. The procedure follows the same steps as in zero-shot classification. Images and texts are encoded, and the dot product between their normalized embeddings at the projection layer is computed. Retrieval performance is evaluated using the recall metric, which measures how often the model retrieves the correct match. It is reported as $\text{recall}@k$, indicating whether the correct image or text appears among the top- k ranked results.

Since zero-shot image classification and zero-shot image-text retrieval are evaluated with the same procedure in CLIP-like models, the retrieval task can be seen as a more fine-grained version of the classification task. While class labels in classification provide only limited semantic information, the sentence descriptions used in retrieval datasets contain more detail and thus offer a richer semantic signal. Retrieval can therefore be understood as an evaluation of how fine-grained and semantically precise the shared embedding space is.

3.4.3 LINEAR PROBING

The third downstream task we evaluate the models on is linear probing. Linear probing refers to the concept of training a simple linear classifier on top of the representations learned by

a model at a specific layer. In contrast to fine-tuning, the weights of the pretrained model remain fixed. Only the separate classification head, which is a single linear layer, is trained on the training split of the evaluation dataset. After training, the performance is measured on the test split.

Linear probing was introduced by Bengio et al. [AB] as a way to evaluate the quality of internal representations learned by a model. The idea is that if a simple linear classifier can solve the classification task well using the model’s features at a given layer, then the features at that layer must already contain useful and linearly separable information for the task. Since the classifier has no activation function and very limited capacity, it can only succeed if the structure of the learned representation already supports the task.

Compared to fine-tuning, linear probing offers a more controlled and interpretable evaluation. Fine-tuning changes the model weights during training on the new task, which makes it harder to isolate whether a model originally learned good general-purpose features. Linear probing, in contrast, gives a clearer picture of how well the original representations transfer to new tasks without interference from further optimization. This makes it especially useful when comparing different model pretraining strategies or architectural choices.

3.4.4 CENTERED KERNEL ALIGNMENT (CKA)

Apart from evaluating the quality of representations different models have learned by testing their usefulness on downstream tasks, one can directly compare them using mathematical methods. This can be done for a single model, by comparing representations learned at different layers, or between models, by comparing representations at the same layer or even between all pairs of layers. One such method is Centered Kernel Alignment.

Centered Kernel Alignment (CKA) is a method specifically developed with the goal of comparing neural network representations. [KNLH] Earlier approaches at comparing representations have utilized methods developed in the context of neuroscience, like Representational Similarity Analysis (RSA) [KMB08], where the goal was to compare the structure of the representational space between different brain regions in a single subject, or the same brain regions in different subjects, even across different species. RSA is also used to compare representations in computational models to representations of stimuli in biological brains.

In contrast, CKA was designed for comparing representations in artificial neural networks, and as such fulfills certain properties. It is invariant to isotropic scaling and orthogonal transformations of the representation matrices, which commonly occurs across training runs with different random seeds, or between models with the same architecture. CKA has been used to analyze differences in representational structure between Vision Transformers and convo-

3 Experimental design

lutional neural networks, as well as the effect of width and depth in neural networks on the learned representations. [RUK⁺, NRK]

Let $X \in \mathbb{R}^{n \times d_1}$ and $Y \in \mathbb{R}^{n \times d_2}$ be representation matrices containing n examples and d_1 , d_2 features, respectively. The matrices are centered across the feature dimension by subtracting the mean of each column:

$$\tilde{X} = X - \frac{1}{n} \mathbf{1} \mathbf{1}^\top X, \quad \tilde{Y} = Y - \frac{1}{n} \mathbf{1} \mathbf{1}^\top Y$$

The *Hilbert-Schmidt Independence Criterion (HSIC)* is then defined as

$$\text{HSIC}(X, Y) = \|\tilde{X}^\top \tilde{Y}\|_F^2$$

where $\|\cdot\|_F$ denotes the Frobenius norm. This computes the sum of squared dot products between all pairs of features from X and Y , yielding a scalar that reflects the overall alignment between the two feature spaces.

To remove the influence of feature scale, CKA normalizes HSIC by the product of the self-similarities:

$$\text{CKA}(X, Y) = \frac{\text{HSIC}(X, Y)}{\sqrt{\text{HSIC}(X, X) \cdot \text{HSIC}(Y, Y)}}$$

This normalization induces invariance to isotropic scaling and ensures that the final similarity score reflects only the directional alignment of the representations.

3.4.5 TEST DATASETS

We evaluate downstream performance and representation quality of the trained models on a total of 15 datasets: 12 for image classification and 3 for image-text retrieval.

The classification datasets include ImageNet, Caltech101, Caltech256, CIFAR10, CIFAR100, Describable Textures Dataset (DTD), FGVC Aircraft, Food101, Oxford-IIIT Pet, Places365, Stanford Cars, and STL10. These benchmarks vary in size, difficulty, and domain, offering a diverse test bed to assess generalization capabilities across different visual categories.

All 12 classification datasets are used in both zero-shot image classification and linear probing. For efficiency in linear probing, we use a smaller subset of ImageNet containing only 100 classes, and further split to two subsets containing only 10% and 1% of the filtered data.

To measure the quality of the shared multimodal representation space, we further evaluate the models on three image-text retrieval benchmarks: Flickr8k, Flickr30k, and MS COCO Captions. These datasets contain images paired with multiple human-written captions and are used for zero-shot retrieval tasks in both directions: image-to-text and text-to-image.

GENERAL OBJECT VS. FINE-GRAINED DATASETS

We group the classification datasets used in this study into two categories: general object and fine-grained datasets. General object datasets consist of broad and commonly occurring categories, such as animals, vehicles, or household objects, that are likely to be reflected in the pre-training distribution. In contrast, fine-grained datasets contain more specific and domain-restricted categories, such as different car models, aircraft types, dog breeds, or textures. This distinction is particularly important in the context of zero-shot classification, since no fine-tuning is performed on the evaluation datasets. The model’s ability to generalize to unseen classes is likely influenced by how closely the test categories align with the data seen during pre-training. [RKH⁺] To account for this, we report average performance scores separately for general object and fine-grained datasets in the zero-shot evaluation.

Table 3.3: Overview of classification datasets used for evaluation

Dataset	Type	Number of Classes	Categories (Examples)
ImageNet (Zero-Shot)	General object	1000	Broad object categories (dog, keyboard, mushroom)
ImageNet-100 (Linear probe)	General object	100	Broad object categories (dog, keyboard, mushroom)
Caltech101	General object	102	Objects and animals (airplane, elephant, face)
Caltech256	General object	257	Objects and scenes (car tire, washing machine)
CIFAR10	General object	10	General object categories (truck, cat, airplane)
CIFAR100	General object	100	Fine-grained objects (maple tree, lawn mower)
STL10	General object	10	Coarse object categories (airplane, truck, deer)
DTD	Fine-grained	47	Visual textures (bubbly, cracked, striped)
FGVC Aircraft	Fine-grained	100	Aircraft models (Boeing 747, Airbus A320)
Food101	Fine-grained	101	Food categories (pizza, pad thai, guacamole)
Oxford-IIIT Pet	Fine-grained	37	Cat and dog breeds (Persian, Pug)
Places365	Fine-grained	365	Indoor and outdoor scenes (kitchen, train station)
Stanford Cars	Fine-grained	196	Car makes and models (BMW M3, Honda Civic)

Table 3.4: Overview of image-text retrieval datasets used for evaluation

Dataset	Number of Images	Captions per Image	Description
Flickr8k	8,000	5	Everyday scenes and activities collected from Flickr
Flickr30k	31,000	5	People-centered scenes from Flickr with diverse actions and objects
MS COCO Captions	123,287	5	Common objects in context with everyday scenes

3.5 IMPLEMENTATION DETAILS

All models are trained using different combinations of our four chosen loss functions. See Table 3.5 for all trained combinations. Training runs for a maximum of 100 epochs, with early stopping applied when the validation loss does not improve for 5 consecutive epochs. To ensure consistency and reproducibility, we fix all random seeds.

3 Experimental design

We use a batch size of 2048, except for models that include the MLM objective, where the batch size is reduced to 1024 due to memory constraints. RandAugment is applied to all training images with `num_transforms=7` and `magnitude=9`.

Training is performed on two NVIDIA A100-SXM4-80GB GPUs using PyTorch Lightning with the DDP strategy. Mixed precision training is enabled with `precision=16-mixed`. Additionally, `torch.set_float32_matmul_precision("medium")` is set to improve performance in matrix multiplications. We enable gradient checkpointing to reduce memory usage during training, and gather batches across devices to ensure consistent contrastive supervision in multi-GPU setups.

Optimization is done with the AdamW optimizer, using a learning rate of $2e-5$, weight decay of 0.1, and betas of (0.9, 0.95). A cosine decay learning rate schedule is used, with a linear warmup during the first epoch, starting from an initial learning rate of $1e-8$. To stabilize training, we apply gradient clipping with a maximum norm of 1.0.

Model checkpoints are saved every 5 epochs. The last checkpoint is always kept, and the best model is selected based on the validation loss, which is computed as the sum of all utilized loss components, divided by the number of losses active in the current training run.

3.5.1 DATASET ABLATION

To investigate how the amount of training data affects the quality of learned representations and downstream task performance, we conduct a dataset ablation study. For this, we train selected model variants on subsets of the full pre-training dataset, using 5%, 10%, 20%, 40%, 60%, 80%, and 100% of the available data.

The subsets are created by randomly sampling from the full dataset without replacement, using a fixed seed to ensure reproducibility. All other training settings, including model architecture, optimizer configuration, and loss functions, remain identical to the full-data training setup.

Due to time constraints, models that include the MLM loss were only trained on fractions up to 40%. We focused on this range because initial experiments with non-MLM models showed the most interesting trends in representation quality and downstream performance within the lower fraction range.

This ablation allows us to evaluate the data efficiency of different loss combinations and to identify how early performance gains can be achieved with reduced data. Results are analyzed in detail in the following chapters.

Table 3.5: Overview of trained models, loss functions, and dataset fractions

Model Name	Loss Functions	Fractions (%)
CLIP	Contrastive	5, 10, 20, 40, 60, 80, 100
CLIP + ITM	Contrastive, ITM	5, 10, 20, 40, 60, 80, 100
CLIP + SimCLR	Contrastive, SimCLR	5, 10, 20, 40, 60, 80, 100
CLIP + SimCLR + ITM	Contrastive, SimCLR, ITM	5, 10, 20, 40, 60, 80, 100
CLIP + MLM*	Contrastive, MLM	5, 10, 20, 40, 100
CLIP + MLM + ITM*	Contrastive, MLM, ITM	5, 10, 20, 40, 100
CLIP + SimCLR + MLM*	Contrastive, SimCLR, MLM	5, 10, 20, 40, 100
CLIP + SimCLR + MLM + ITM*	Contrastive, SimCLR, MLM, ITM	5, 10, 20, 40, 100
SimCLR	SimCLR	5, 10, 20, 40, 60, 80, 100
SimCLR + ITM	SimCLR, ITM	5, 10, 20, 40, 60, 80, 100
MLM + ITM*	MLM, ITM	100
SimCLR + MLM + ITM*	SimCLR, MLM, ITM	5, 10, 20, 40, 100

*Models including MLM are trained only on dataset fractions up to 40% and use a batch size of 1024 instead of 2048 due to memory constraints.

Table 3.6: Number of training samples used per dataset fraction

Fraction (%)	Number of Samples
5	197,500
10	395,000
20	790,000
40	1,580,000
60	2,370,000
80	3,160,000
100	3,950,000

3.5.2 EVALUATION DETAILS

We use the datasets listed in Table 3.3 for evaluation on zero-shot classification and linear probing. For image-text retrieval, we use the datasets listed in Table 3.4.

To construct the text-based classifier in zero-shot classification, we encode class names using different prompt templates. Since previous work has shown that the choice of prompt templates can impact performance, we evaluate accuracy across multiple template sets. [RKH⁺] Table 3.7 lists the individual template sets, which are taken from the official CLIP GitHub repository.¹

¹<https://github.com/openai/CLIP/blob/main/data/prompts.md>

Template Set	Templates
1 (0 templates)	[""]
2 (1 template)	["a photo of a ."]
3 (3 templates)	["a photo of a .", "a photo of a small .", "a photo of a big ."]
4 (5 templates)	Template Set 3 + ["a bad photo of a .", "a good photo of a ."]
5 (9 templates)	Template Set 4 + ["a blurry photo of a .", "a black and white photo of a .", "a low contrast photo of a .", "a high contrast photo of a ."]
6 (18 templates)	Template Set 5 + all templates from Template Set 5 but article "a" is replaced with "the" (as in "a photo of the .")

Table 3.7: Prompt template sets used for zero-shot classification.

For linear probing, we train a set of linear classifiers in parallel on the CLS token from the final encoder layer, using a range of learning rates: 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.3. All probes are trained for a maximum of 100 epochs using stochastic gradient descent (SGD). We report the result for the best-performing probe based on downstream accuracy.

The batch size for all evaluation tasks is set to 128.

4 RESULTS

In this chapter, we will first present and discuss results of our experiments in order, followed by a summary discussion focused on each loss function individually. We will start with the results of the zero-shot tasks for image classification and image-text retrieval, followed by linear probing. We then present a short section on model similarity analysis with Centered Kernel Alignment, before concluding this chapter with a discussion of each loss.

4.1 ZERO-SHOT IMAGE CLASSIFICATION

For zero-shot image classification, we only evaluate models trained with CLIP [RKH⁺], as this loss function enables the zero-shot capability through the learned alignment between image and text vectors in the shared embedding space. Our experiments validated that models trained without CLIP don't have this zero-shot capability, as evidenced by reduced classification accuracy. See Appendix 5.2 for full results.

PROMPT TEMPLATE ABLATION

We start our exposition of the zero-shot classification results with an ablation over different sets of prompt templates for constructing the text-based classifier, since we have found pronounced differences between models depending on the choice of templates. An explanation of the idea behind prompt templates is given in section 3.4.1. This table 3.7 lists the templates for the different sets.

We find that all but one model see a significant drop in accuracy when using a single prompt template rather than embedding the class name as is, without a prompt. We also find that CLIP and CLIP+MLM models can close this drop in accuracy when using more templates, while CLIP+SimCLR+ITM models trained without MLM can't close this gap and achieve highest zero-shot accuracy without using templates.

Table 4.1 lists the Top-1 accuracies for every model for every set of templates.¹ Scores are averaged over the three groups of datasets, please see section 3.4.5 for an explanation of the

¹Tables with results for all datasets for every set of templates are included in the Appendix 5.2.

4 Results

Templates	All Datasets						General Object						Fine-Grained					
	<i>General object + Fine-grained</i>						<i>ImageNet, CIFAR-10, CIFAR-100, Caltech-101, Caltech-256, STL10</i>						<i>Foods, Pets, Aircrafts, Cars, Textures, Places365</i>					
	0	1	3	5	9	18	0	1	3	5	9	18	0	1	3	5	9	18
CLIP+MLM+ITM	25.47	26.17	26.48	26.54	26.88	26.91	41.56	42.75	43.26	43.39	43.79	43.54	9.39	9.59	9.70	9.70	9.97	10.29
CLIP+MLM	25.61	24.14	25.10	25.32	26.62	26.53	41.28	38.84	40.54	41.00	42.89	42.72	9.93	9.44	9.65	9.64	10.36	10.34
CLIP	25.37	22.43	25.03	25.26	26.37	26.00	40.97	36.14	40.67	41.09	42.24	41.50	9.76	8.72	9.38	9.43	10.49	10.51
CLIP+SimCLR+MLM	24.95	23.37	24.87	25.09	26.02	25.98	39.69	36.96	39.36	39.65	41.13	41.14	10.21	9.78	10.39	10.52	10.90	10.83
CLIP+SimCLR+MLM+ITM	24.84	22.33	23.06	23.92	25.26	24.84	39.34	34.87	36.53	38.13	40.12	39.45	10.34	9.79	9.59	9.70	10.41	10.23
CLIP+ITM	25.22	21.80	23.58	23.68	25.07	24.49	41.28	34.80	37.52	37.54	39.72	38.81	9.15	8.79	9.64	9.83	10.43	10.16
CLIP+SimCLR+ITM	23.98	19.18	20.79	21.17	22.58	22.44	38.40	30.04	33.05	33.45	35.84	35.68	9.55	8.32	8.52	8.88	9.33	9.21
CLIP+SimCLR	23.57	19.97	21.85	22.18	22.82	22.68	37.78	31.48	34.45	35.11	35.54	35.34	9.36	8.46	9.26	9.24	10.10	10.02

Table 4.1: Zero-shot image classification: Top-1 accuracy (%) for each set of prompt templates.

Results are averaged across all datasets, general object datasets, and fine-grained datasets. Models are split in 3 groups: CLIP+MLM+ITM sees steady increase in accuracy with more templates. Rest of MLM models and CLIP decrease from 0 to 1 templates, but have highest accuracy at 9 templates. Models without MLM have highest accuracy with 0 templates. Green marks the best value in each group, blue the second best, and red the lowest. Models also ordered by highest achieved accuracy.

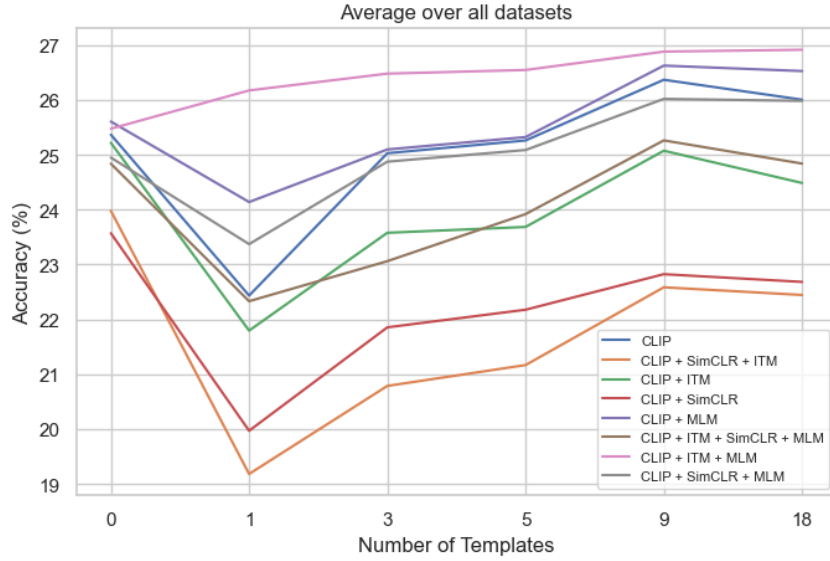


Figure 4.1: Zero-shot image classification: Top-1 accuracy as a function of number of templates.

Scores averaged over all tested datasets. Only CLIP+MLM+ITM doesn't have lowest accuracy with only one template.

grouping. Figure 4.1 plots the accuracies as a function of the template number for better visualization.

The Top-1 accuracy for our models ranges between 19% and 27% averaged over all evaluated datasets, depending on the chosen set of templates.

As discussed in section 3.4.1, since performance in zero-shot classification is dependent on the size and distribution of the pre-training dataset, we can observe more significant differences in model performance on datasets of the general object category than the fine-grained category. For the general object category of datasets, average Top-1 accuracy ranges between 30% and 43% (depending on template set), while average Top-1 accuracy on the fine-grained datasets varies less widely between models, in the 8% to 10% range.

NO TEMPLATE IS MOSTLY BETTER THAN A SINGLE TEMPLATE Contrary to results reported in the literature [RKH⁺], we find that embedding class names as a plain text "{class name}" without a prompt achieves better accuracy for most models than embedding class names in the simple prompt "a photo of a {class name}.". Switching from no template to the single template reduces accuracy significantly for most models.

Average accuracy in the general object category drops between 3% and 8% between all models, with the highest drop for model CLIP+SimCLR+ITM with more than 8% and the lowest drop for models CLIP+MLM and CLIP+SimCLR+MLM between 2%-3%.

CLIP models trained with the MLM loss, as well as just CLIP are the most robust to this change, while the other models with CLIP, SimCLR and ITM are the least robust.

The only exception to this behavior is CLIP+MLM+ITM, which increases by more than 1% when using a single template rather than no template.

MORE TEMPLATES ONLY HELP MODELS CLIP AND CLIP+MLM Some models achieve highest average accuracy when using no prompt, others when using a set of nine prompts. Specifically, model CLIP and models that were trained with CLIP+MLM achieve best average accuracy with nine prompts, while models CLIP+ITM, CLIP+SimCLR, CLIP+SimCLR+ITM, which were trained without MLM, report highest average accuracy without using a prompt.

Only model CLIP+MLM+ITM doesn't see this drop in accuracy going from zero to one template, and also sees a steady increase in accuracy as we add more templates, achieving highest accuracy with eighteen templates.

REDUNDANT TEMPLATES REDUCE ACCURACY Another finding is that classification accuracy decreases for most models when going from a prompt set with nine templates to a

prompt set with eighteen templates. We suspect that this is due to the nature of the added prompts: the set with eighteen templates simply duplicates the nine templates of the previous set, but replaces the article "a" with "the". For example, template "a photo of a black {}" becomes "a photo of the black {}". We hypothesize that this addition of templates introduces more noise than signal, as the added templates are redundant because they don't provide more meaningful semantic diversity, and therefore lead to a decrease in classification accuracy.

BEST ACHIEVED RESULTS

Table 4.2: **Zero-shot classification: Highest achieved accuracy for each model, depending on template set.** Scores for averages across all datasets, general object datasets, and fine-grained datasets. Best values per dataset group are highlighted in green.

Model	All Datasets	General Object	Fine-Grained
CLIP+MLM+ITM	26.91	43.54	10.29
CLIP+MLM	26.62	42.89	10.36
CLIP	26.37	42.24	10.51
CLIP+SimCLR+MLM	26.02	41.14	10.90
CLIP+SimCLR+MLM+ITM	25.26	40.12	10.41
CLIP+ITM	25.22	41.28	10.43
CLIP+SimCLR+ITM	23.98	38.40	9.55
CLIP+SimCLR	23.57	37.78	10.10

We can select the best achieved average accuracy for every model across all evaluated template sets, which are listed in Table 4.2.

CLIP without any additional loss function ranks in third place. Adding MLM as well as MLM+ITM improves performance on the general object datasets by 0.65% and 1.3% respectively. All other models show worse performance.

The addition of SimCLR or ITM to CLIP significantly decreases accuracy by a couple percent, while adding MLM on top of that is able to mitigate the negative impact.

CHANGE COMPARED TO CLIP

For an extended visualization, Figure 4.2 plots the change in accuracy for all models compared to the CLIP baseline, as a function of the used template set. Only models CLIP+MLM and CLIP+MLM+ITM improve accuracy compared to CLIP over all evaluated template sets, while all other models generally show worse accuracy, with the exception of CLIP+SimCLR+MLM which has 1% better accuracy to CLIP when using one template.

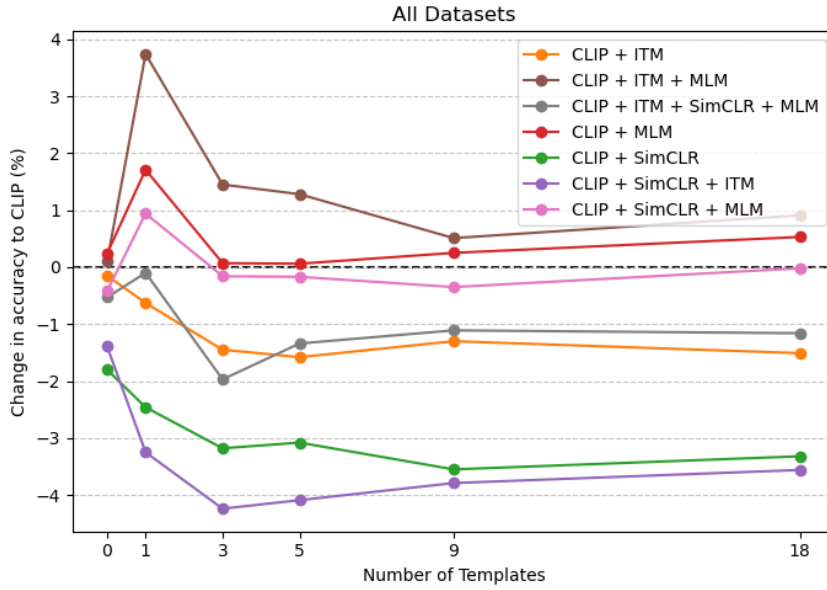


Figure 4.2: **Zero-shot classification: Change in Top-1 accuracy compared to CLIP as a function of template amount.** Scores are averaged across all tested datasets.

MASKED LANGUAGE MODELLING BENEFITS ZERO-SHOT CLASSIFICATION The addition of MLM to CLIP boosts zero-shot accuracy, but only in the absence of losses SimCLR and ITM. While these loss functions degrade performance, the added supervision on the language stream leads to better text representations with an improvement in zero-shot classification.

UNIQUE EFFECT OF CLIP+MLM+ITM Model CLIP+MLM+ITM displays a unique performance in our setup as the only model whose classification accuracy on the zero-shot task rises with more added prompt templates, compared to the decrease we observe when going from the promptless approach to using prompt templates.

As will be seen in the next section on zero-shot image-text retrieval, this model performs worse in the retrieval task than CLIP+MLM, while it performs better than CLIP+MLM in the classification task, although the difference in accuracy is only 0.6% averaged over the general object datasets.

These results suggest that the ITM loss in combination with MLM is not able to capture the fine-grained alignment that’s necessary in the retrieval task, but is able to drastically improve robustness in zero-shot classification.

SIMCLR AND ITM DEGRADE ZERO-SHOT PERFORMANCE While the ITM loss brings a unique benefit in combination with CLIP and MLM, it does not increase zero-shot performance in any other combination. The same holds for the SimCLR loss. As will be seen throughout the experiments, and discussed at the end of this chapter, losses SimCLR and ITM generally decrease performance on the downstream tasks. For SimCLR, this is most likely a consequence of it’s application to the same layer as the CLIP loss, which leads to interference in learning of multimodal representations capable of good zero-shot performance. The same can be said about ITM, although to a lesser degree.

IMAGE-TEXT MATCHING HEAD IN ZERO-SHOT CLASSIFICATION

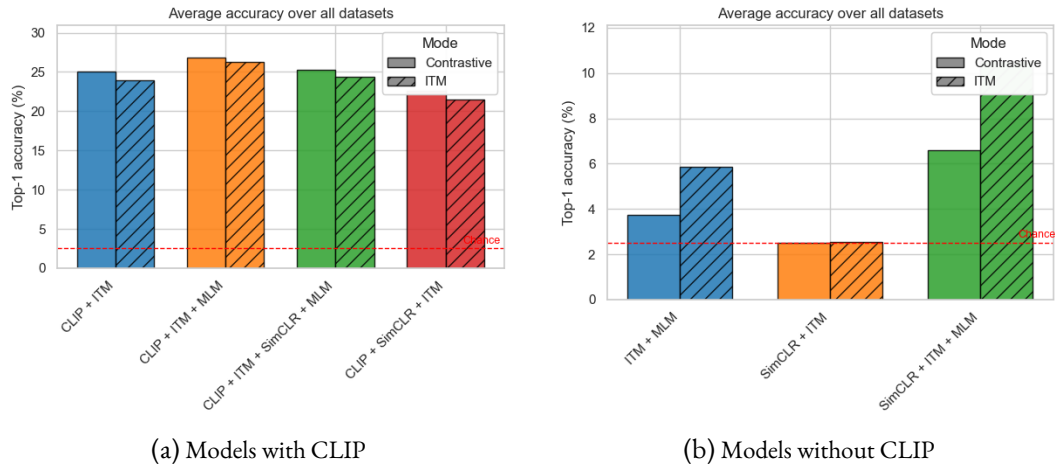


Figure 4.3: **Zero-shot classification accuracy: Using cosine similarity (regular) vs. ITM head (ITM) for final prediction.**

The results so far were obtained by choosing the predictions for classification with the standard method utilized in CLIP: selection of the class with the highest cosine similarity to an image in the multimodal embedding space. Another way to obtain a prediction is by using the matching head for models that were trained with Image-Text Matching. Figure 4.3 plots the difference in accuracy when using the ITM head for the final prediction. Subplots are for CLIP and non-CLIP models.

For this method, we choose a prediction by selecting the top-k most similar classes according to cosine similarity at the final embedding layer, then pairing each class embedding with the image embedding, and running the pairs through the image-text matching head to obtain the final prediction.

For models trained with CLIP and ITM, we observe that using the ITM head does not lead to better zero-shot performance, as classification accuracy slightly decreases by 1-2%. This

shows us that training with CLIP is enough to learn good multimodal features. Since utilizing the ITM head for the final prediction leads to worse accuracy, and since models trained with CLIP and ITM perform worse on average than models trained with just CLIP, we can also conclude that adding our version of ITM to CLIP is not beneficial in the zero-shot classification task.

We also evaluate results for models trained with ITM but without CLIP. For these models, using the ITM head improves the classification accuracy on two out of three models. In particular, models SimCLR+MLM+ITM and MLM+ITM see about 4% and 2% increase in accuracy respectively, while Model SimCLR+ITM doesn't see an increase. It's performance stays at chance level, showing that it failed to learn any meaningful features. Models SimCLR+MLM+ITM and MLM+ITM perform a couple percent above chance level, but have significantly lower classification accuracy than the CLIP models.

In conclusion, these results show us that the ITM loss used in our study fails to learn sufficient multimodal representations that are suitable for zero-shot classification.

4.2 ZERO-SHOT IMAGE-TEXT RETRIEVAL

Model	Image Retrieval (Recall@1)				Text Retrieval (Recall@1)			
	Flickr30k	Flickr8k	MSCOCO	Average	Flickr30k	Flickr8k	MSCOCO	Average
CLIP+MLM	35.84%	35.14%	25.16%	32.05%	51.10%	51.30%	38.90%	47.10%
CLIP+SimCLR+MLM+ITM	34.88%	34.28%	24.50%	31.22%	51.60%	51.50%	38.46%	47.19%
CLIP+SimCLR+MLM	33.24%	33.80%	22.25%	29.76%	48.30%	50.20%	35.84%	44.78%
CLIP+MLM+ITM	32.18%	32.66%	21.26%	28.70%	46.80%	47.30%	33.06%	42.39%
CLIP	30.20%	30.12%	22.72%	27.68%	45.60%	45.70%	35.48%	42.26%
CLIP+ITM	28.80%	31.12%	22.02%	27.31%	43.60%	45.30%	34.96%	41.29%
CLIP+SimCLR	28.68%	29.96%	21.25%	26.63%	44.60%	43.60%	34.96%	41.05%
CLIP+SimCLR+ITM	28.64%	29.36%	21.12%	26.37%	42.70%	43.10%	33.92%	39.91%

Table 4.3: Zero-shot retrieval: Recall@1 for Image-Text Retrieval across Flickr30k, Flickr8k, and MSCOCO datasets (in %). Best scores per dataset/average highlighted in bold.

Table 4.3 shows the percentage scores for Recall@1 for Image Retrieval and Text Retrieval, and figure 4.4 plots the change in accuracy to the CLIP baseline for every model.

MLM IMPROVES RETRIEVAL IN ANY CONFIGURATION We observe that in contrast to zero-shot classification, adding the masked language loss improves recall performance in both image and text retrieval tasks in any configuration. While the addition of MLM alone has the most benefit, improving recall@1 over the CLIP baseline by 4.36% in image retrieval and 4.92% in text retrieval, adding MLM in any other combination with the SimCLR and ITM losses also improves over the baseline. In the zero-shot classification task, adding SimCLR or

4 Results

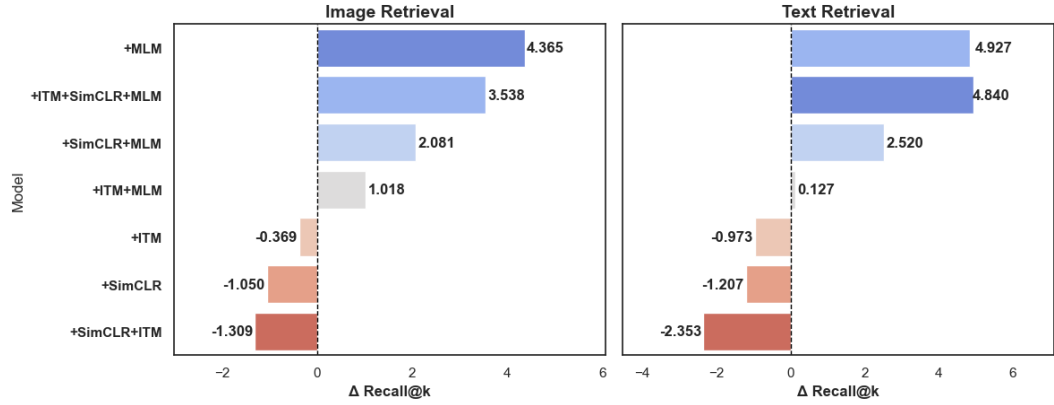


Figure 4.4: **Zero-shot retrieval: Change in Recall@1 to CLIP baseline.** Loss additions are indicated on left hand side of subplots.

SimCLR+ITM to CLIP+MLM decreased the model’s performance, a change that we also observe in this task. Contrary to the result in classification, this decrease doesn’t drop the models performance below that of the baseline set by CLIP. This suggests that adding MLM has a generally stronger effect in retrieval than in classification. This could be because text labels in the retrieval datasets are more fine-grained, as they are sentence descriptions that contain more semantic signal, which would be benefitted more by additional training signal of the language loss.

SIMCLR AND ITM DECREASE RETRIEVAL PERFORMANCE We observe that adding SimCLR to the loss setup generally decreases the retrieval performance. Adding SimCLR to CLIP alone reduces recall by about 1% in image retrieval and 2.3% in text retrieval. Adding SimCLR to CLIP+MLM reduces both image and text retrieval by more than 2%. Similarly, adding SimCLR to CLIP+ITM reduces image and text retrieval by about 1%.

While we saw a unique benefit of model CLIP+MLM+ITM in the zero-shot classification task compared to all other models, we can see a reversal of this effect in the retrieval task. Adding ITM to CLIP+MLM decreases recall by more than 3% in image retrieval and almost 5% in text retrieval. Based on this observation, we hypothesize that the ITM head doesn’t seem to be able to capture the fine-grained alignment necessary in the retrieval task, but is able to capture broader features that could be more beneficial in zero-shot classification, but only in combination with the masked language modelling loss.

4.3 LINEAR PROBING

After testing our models on the zero-shot tasks, which depends on the alignment between image and text embeddings in the shared multimodal vector space, we will now turn to an evaluation of visual representations. To do this, we test accuracy on linear probes which were trained on the [CLS] token from the last layer of the image encoder. We test on the same datasets as in zero-shot classification, with a slight adjustment for ImageNet. For ImageNet, we test only on 100 out of the 1000 classes, and construct two splits respectively containing only 1% and 10% of the filtered dataset, for reasons of training efficiency.

For this evaluation, we now also include the models that were trained without the CLIP loss, namely models SimCLR, SimCLR+MLM, ITM+MLM and SimCLR+ITM+MLM.

Model	Average (All)	Average (General object)	Average (Fine-grained)
CLIP+MLM	67.83	79.90	53.75
CLIP	67.05	78.94	53.19
CLIP+ITM	66.96	78.95	52.97
CLIP+ITM+MLM	66.60	78.76	52.42
CLIP+SimCLR+ITM	64.60	76.89	50.27
CLIP+ITM+SimCLR+MLM	65.95	78.49	51.31
CLIP+SimCLR+MLM	65.38	77.90	50.76
CLIP+SimCLR	64.59	76.65	50.52
SimCLR+ITM+MLM	52.81	65.05	38.53
SimCLR+ITM	43.01	54.27	29.88
SimCLR	42.71	53.90	29.65
ITM+MLM	32.09	43.44	18.84

Table 4.4: **Linear probing: Top-1 accuracy.** Scores averaged over all datasets, general object datasets, and fine-grained datasets. Models are separated by CLIP vs non-CLIP.

Table 4.4 shows the Top-1 accuracy for each model, averaged over all datasets, over the general object datasets and the fine-grained datasets. As the linear probes are fine-tuned on the tested dataset, we observe much higher accuracies than in zero-shot classification, especially on the fine-grained datasets.

CLIP MODELS OUTPERFORM NON-CLIP MODELS Models trained without CLIP perform significantly worse in linear probing than models trained with CLIP, being between 15-30% worse averaged over all datasets. These results especially highlight the insufficiency of the Image-Text Matching loss in learning good features, as model ITM+MLM has the worst linear probe accuracy with 32.09% averaged over all tested datasets. Models trained with SimCLR achieve about 15% more accuracy, and the addition of ITM to the SimCLR model does not lead to a significant increase in performance. On the other hand, model SimCLR+ITM+MLM

sees an additional increase of almost 10% compared to SimCLR/SimCLR+ITM. While these results show us that the ITM loss on it's own does not lead to sufficient visual representations, the combination of the two unimodal losses with the ITM loss synergizes in a way to learn better representations than just one unimodal loss combined with ITM.

Still, linear probe performance is much worse than for models trained with CLIP, which achieve more than 60% accuracy averaged over all datasets.

CHANGE IN ACCURACY TO CLIP

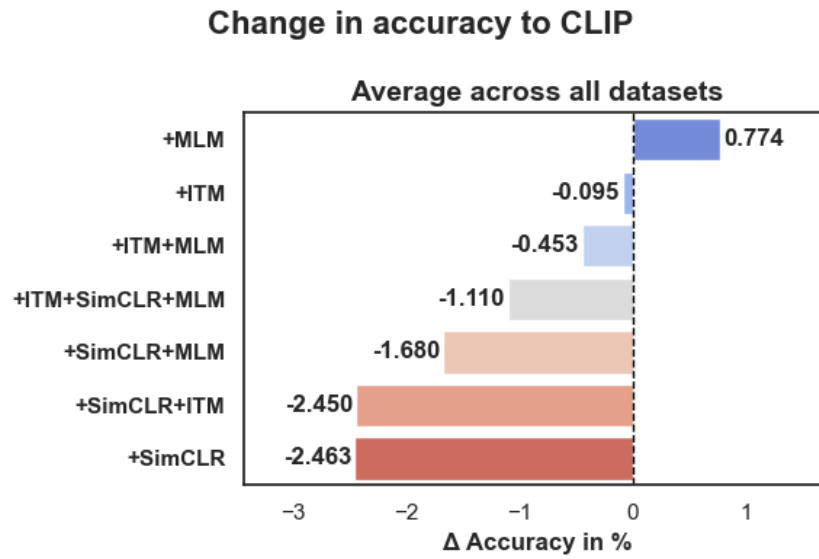


Figure 4.5: **Linear probe: Change in accuracy compared to CLIP.** Scores averaged over all datasets.

Figure 4.5 shows the difference in accuracy of CLIP models with added losses to the CLIP baseline when averaged over all datasets.

MASKED LANGUAGE MODELLING IMPROVES CLIP VISUAL REPRESENTATIONS As can be seen in Figure 4.5, adding MLM improves accuracy of the linear probes compared to the CLIP baseline by about 0.7% when averaged over all tested datasets. While this is only a slight improvement, we can tentatively state that improving the representations learned by the text encoder with the additional text-only loss doesn't just improve the multimodal representations, as evidenced in the zero-shot tasks, but also improves the representations learned by the visual backbone itself. Since the zero-shot tasks depend equally on the representations learned by the visual and text encoders, the benefit of adding MLM is more pronounced in these tasks. In linear probing, the benefit of MLM comes indirectly through improved

alignment in the CLIP loss. As the text encoder learns better representations, convergence in the CLIP loss is improved, also leading to better learned visual representations. Since this improvement is indirect, we only see a slight increase in accuracy compared to zero-shot classification.

While adding any other combination of loss functions to the CLIP baseline decreases linear probe accuracy, we observe that adding MLM to CLIP model decreases the negative effect of the other loss functions. Adding MLM to CLIP+SimCLR decreases the gap to the CLIP baseline from 2.4% to 1.6%, and adding MLM to CLIP+SimCLR+ITM decreases the gap from 2.4% to 1.1%.

SIMCLR DECREASES VISUAL REPRESENTATION QUALITY WHEN COMPETING WITH CLIP The addition of SimCLR to CLIP significantly decreases linear probe accuracy. Adding SimCLR alone has the most negative effect, decreasing linear probe accuracy by 2.4% on average compared to CLIP. The addition of MLM or ITM to CLIP+SimCLR improves performance, showing the overall detrimental effect of the SimCLR loss in our setup.

We hypothesize that the reason for this is the competing effect experienced between the CLIP and SimCLR loss. Both losses are fed with features from the same projection head. This results in a "tug-of-war" between the two loss functions, both trying to optimize the layer output to minimize their respective objectives. The CLIP objective is to align the matching image and text embeddings in the projection space, while the SimCLR objective is to align the embeddings of the two different image views in that same projection space. The effect seems to be a degradation of the general quality of the visual representations learned by the encoder. While the authors of SimCLR found that the representations of the layer before the projection head achieve better downstream task performance [CKNH], this effect doesn't seem to apply in conjunction with CLIP.

CLIP IS SUFFICIENT FOR A GOOD VISUAL BACKBONE The results overall suggest that training with CLIP without any additional losses is enough to learn strong visual representations. While we see an increase in linear probe accuracy when adding MLM to the text encoder, this improvement is only marginal compared to the effects we observe in the zero-shot tasks. Addition of the other losses even proves detrimental. While the negative effect of SimCLR can be explained by the competition with CLIP over gradients in the projection space, adding ITM also provides no benefit, maybe for the same reason. This shows that using CLIP to train a visual backbone with language supervision is sufficient to learn good visual representations, and better than training with self-supervision on images alone.

LINEAR PROBING ACROSS MODEL LAYERS

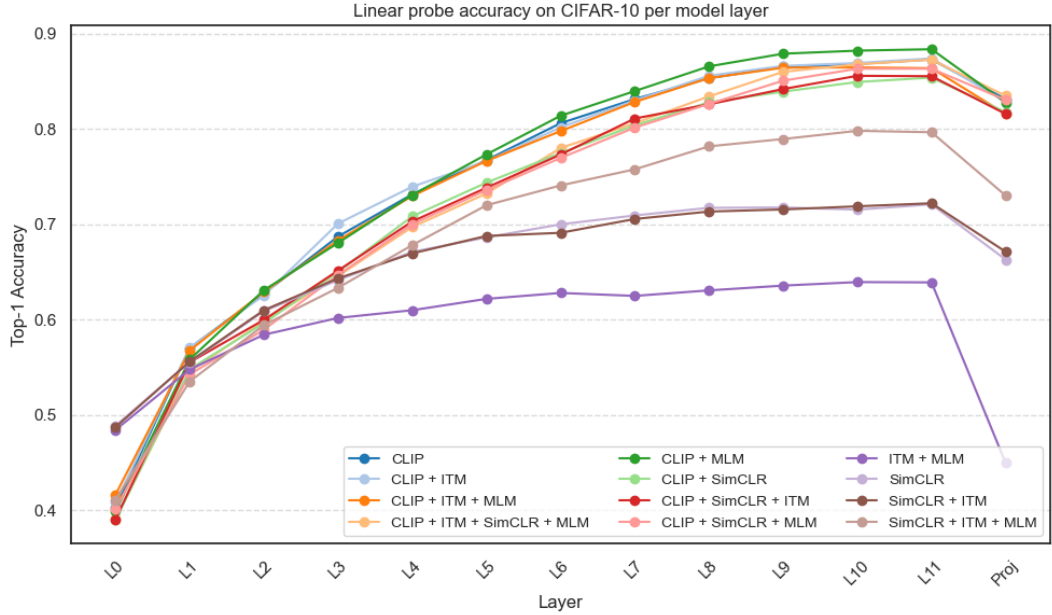


Figure 4.6: **Linear probe: Accuracy on CIFAR-10 per model layer.** "Proj" is the visual projection layer.

In addition to testing linear probes on the last encoder layer, we also test linear probes over all model layers. Figure 4.6 plots the accuracy for each layer, tested on CIFAR-10.

We observe that models trained without CLIP see much less growth in accuracy over the layers, especially model ITM+MLM. While the non-CLIP models have higher linear probe accuracy on the first layer than the CLIP models, this just suggests that they fail to learn good representations at the later layers, as the first layer does not specialize to learn low-level features that the later layers can build upon. CLIP models on the other hand see a steady, logarithmic growth over the layers, suggesting that the layers learn ever more complex features. Model SimCLR+ITM+MLM comes closer to this behavior than the other non-CLIP models, underscoring the potential synergistic effect between these three losses.

Linear probe accuracy significantly decreases for all models going from the last encoder layer to the visual projection layer. Models trained with CLIP and SimCLR apply their loss functions at this layer, and representations at this layer become optimized for the loss objective and thus less useful for general downstream transfer. We see an even bigger drop in accuracy for ITM+MLM. This could come about as the representations are prepared for subsequent processing in the ITM head. Still, the generally low performance of the visual encoder representations suggests that the ITM loss generally fails to learn useful visual features.

We additionally observe that the decline of linear probe accuracy for CLIP+SimCLR models already starts at the third encoder layer, and non-CLIP SimCLR models further decrease in accuracy a few layers deeper. This shows that overall CLIP learns better representations than SimCLR across all models layers, underscoring the effectiveness of language supervision for representation learning compared to self-supervision.

4.4 ABLATION OF DATASET SIZE

This section will go over the results of an ablation experiment on the size of the training dataset, where we trained models of different loss combinations on fractions of the full dataset, ranging from 5% to 100%.

Practice has shown that training models with contrastive loss functions is particularly data-intensive, requiring more data than other training approaches to reach similar performances. [RKH⁺] We thus conduct this ablation experiment to determine potential data efficiency effects in training with different losses. To summarize, we find that there are benefits to training CLIP with additional self-supervision in a low-data regime between 5% and 40% of the dataset across all of our evaluated tasks.

As of the date of publication, because of time constraints we unfortunately were not able to obtain models trained with CLIP and MLM on fractions above 40%. Since during the course of this study we had observed the more interesting results for fractions between 5% and 40%, we decided to prioritize models trained in this range. We hope that future research can close this gap.

Furthermore, all models tested in this section were trained with weaker image augmentation settings in the RandAugment module. Please refer to section 3.2.3 for more details.

While this does change results a bit compared to the models trained on the full dataset evaluated in the last three sections, we think that our findings are still generally valid regardless of the difference in image augmentation strength. A discussion and study of the influence of augmentation strength on CLIP and SimCLR models is held in the discussion section 4.6.

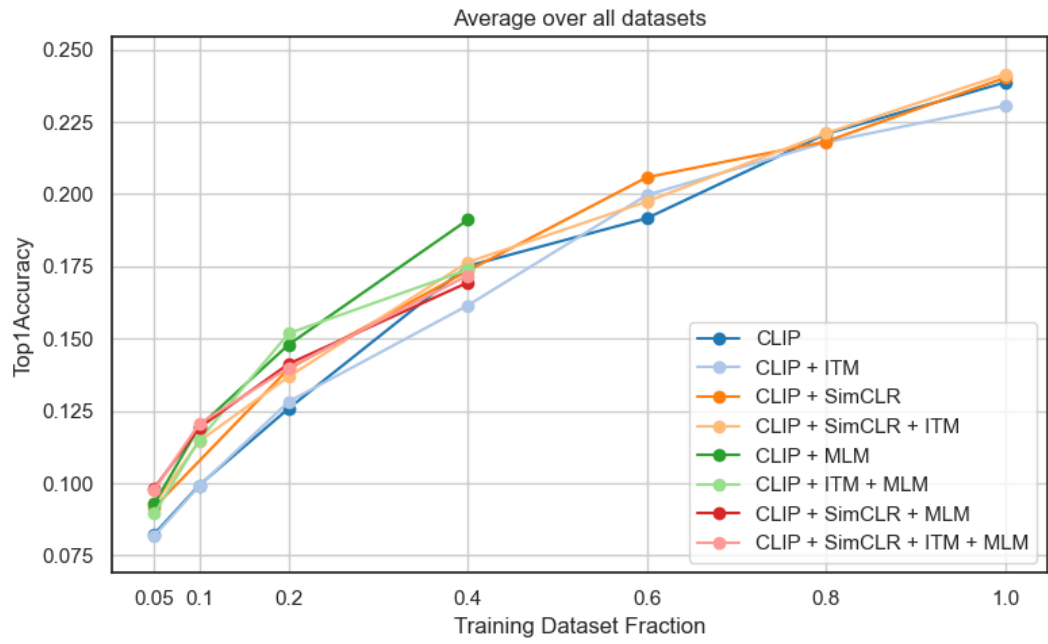
Like in the previous sections, we will start with an evaluation of the zero-shot tasks, followed by linear probing of the visual backbone, all tested on the same datasets.

Table 3.6 in Chapter lists the amount of examples in each fraction of the dataset.

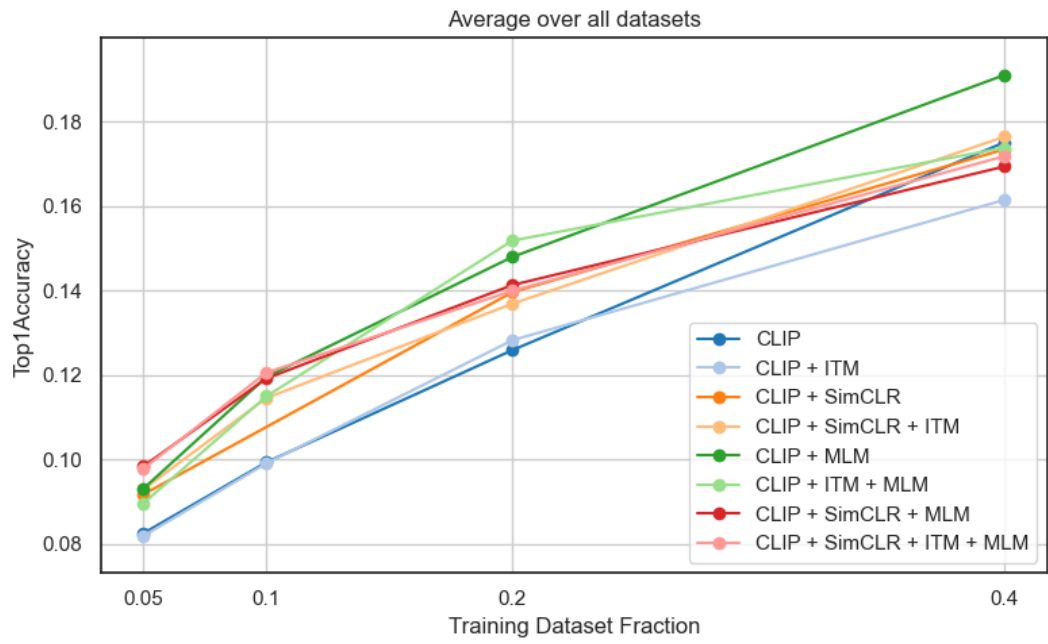
4.4.1 ZERO-SHOT CLASSIFICATION

Figure 4.7 plots the Top-1 accuracy averaged over all datasets for each model as a function of the dataset fraction. The upper subplot shows accuracies over the full range of fractions, the lower subplot shows accuracies focused on the 5% to 40% fraction. To simplify the analysis

4 Results



(a) Fraction: 5% to 100%



(b) Fraction: 5% to 40%

Figure 4.7: Zero-shot classification: Top-1 Accuracy per fraction of training data, averaged over all datasets and all template sets.

and average out the variation we observed for zero-shot classification with different template sets, we take the average model scores over all template sets.

While we observe a general linear trend with slightly logarithmic growth going from the lowest fraction to the full dataset size, it is in the low-data regime between 5% and 40% of training data that we see differences between the models.

Starting at the lowest fraction of training data, we see that models that employ the additional self-supervised losses in either modality achieve better performance than models trained without them (CLIP and CLIP+ITM). This gap in task accuracy stays up to an increase of 40% of full training data. We further see that models using the MLM loss but without the SimCLR loss grow even quicker in task accuracy, especially CLIP+MLM.

We can thus see that only using the CLIP loss needs more training data than the other configurations to learn good multimodal representations, as the performance of CLIP only starts to catch up to the CLIP+SimCLR models once we increase training data to 40% of the full set. In the low-data regime below that fraction, the addition of the self-supervised losses proves beneficial, as these loss functions seem to help the model in extracting more information out of the limited data.

Since we don't have datapoints for MLM models available above the 40% fraction, we can only speculate based on the performance of the models trained on the full dataset that this group of models will see continued and accelerated growth in task accuracy with an increase in training data, ultimately outperforming the other models. We hope future research can test this hypothesis.

4.4.2 ZERO-SHOT RETRIEVAL

We see a similar pattern in the zero-shot retrieval task. Figure 4.8 plots Recall@1 scores in the image- and text-retrieval tasks over the full range of dataset fractions, and in the 5% to 40% dataset fraction

CLIP without any self-supervision shows lower performance than CLIP models with added self-supervision in the 5% to 10% range, with slightly stronger effects for text- than image-retrieval. A bit earlier than in zero-shot classification, this gap starts to close once training data increases to 20% of the full size.

4.4.3 LINEAR PROBING

Figure 4.9 plots the Top-1 accuracy of linear probes trained on the [CLS] token of the last image encoder layer. Score is averaged over all datasets. Here we will once again include models SimCLR, SimCLR+ITM and SimCLR+ITM+MLM. We don't have datapoints for model

4 Results

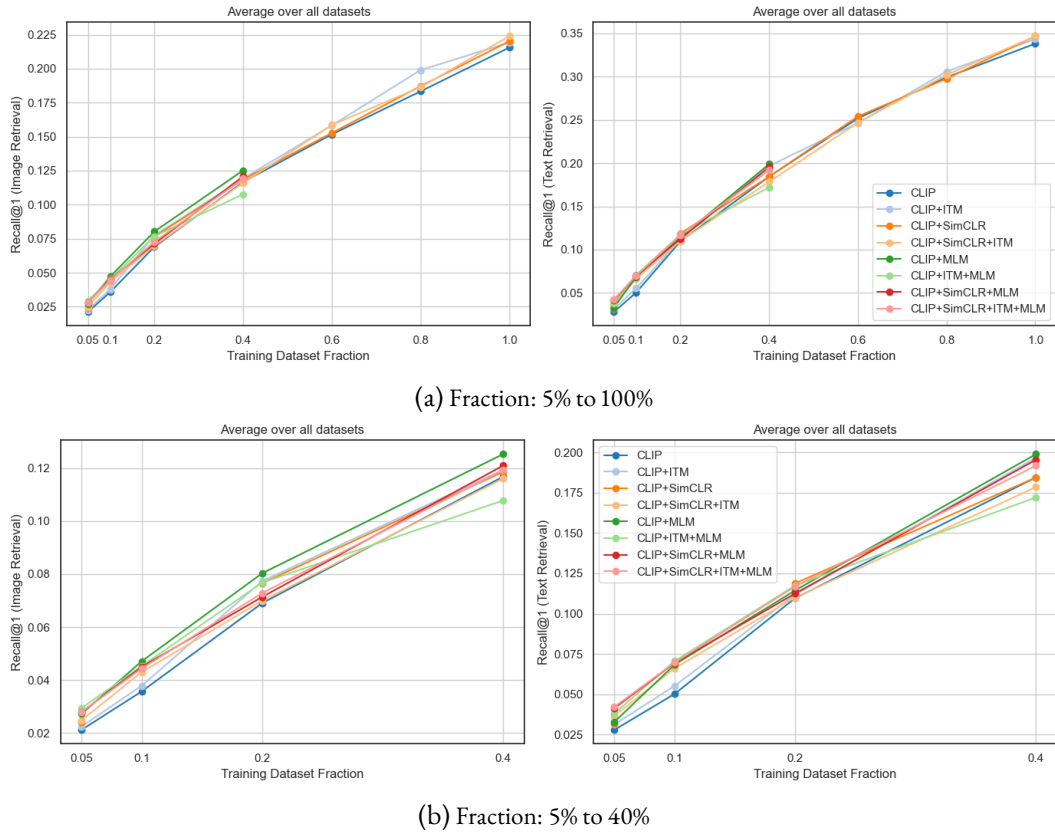


Figure 4.8: Zero-shot image-text retrieval: Recall@1 per fraction of training data, averaged over all datasets.

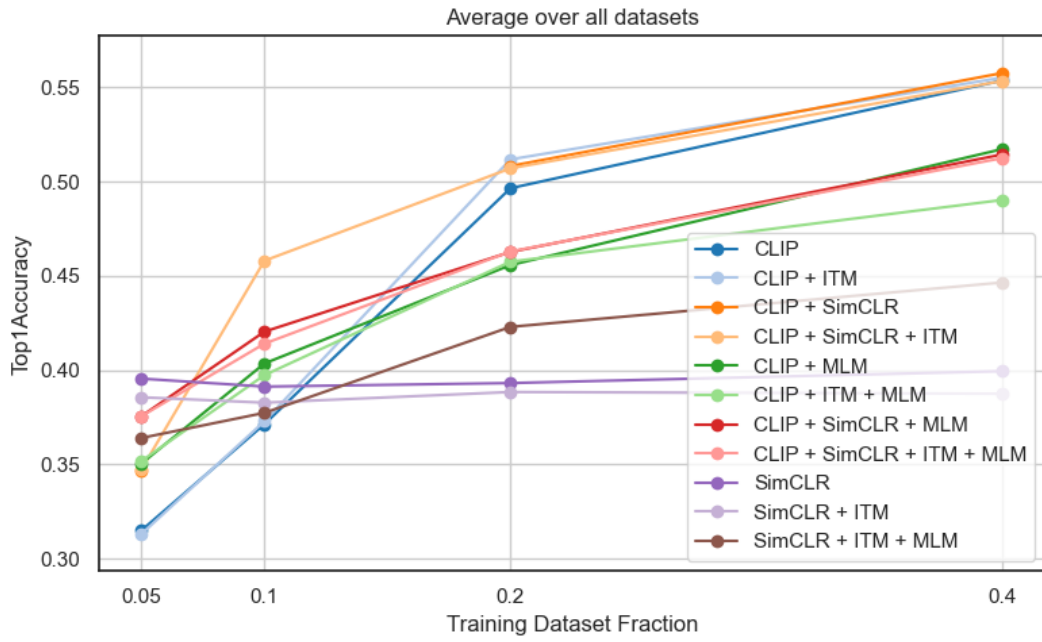
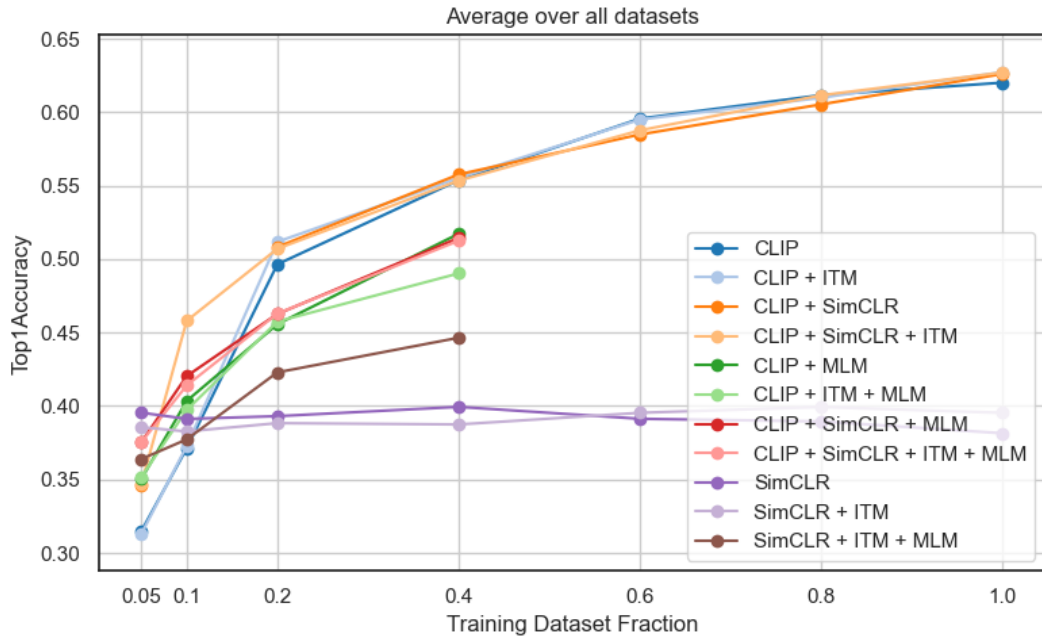


Figure 4.9: Linear probe: Top-1 Accuracy per fraction of training data, averaged over all datasets.

ITM+MLM since we found that it’s task performance compared to the other models didn’t necessitate the ablation study.

SIMCLR SATURATES AT LOW AMOUNT OF DATA We observe that SimCLR (models SimCLR and SimCLR+ITM) doesn’t increase in linear probe accuracy with growing amount of training data. The loss already saturates at just 5% of the training data, which are about 197,500 images, and the addition of more images doesn’t lead to better representations. In practice, we observe that the training and validation loss decreases to near zero after about 2000 steps.

Looking only at models trained with CLIP, we again see the most differences in performance in the low-data regime between 5% and 40%.

Again, models using self-supervision on either modality outperform the pure CLIP model at the 5% and 10% fraction. But just as in the zero-shot tasks, we see a big jump in accuracy when increasing the training data fraction to 20%, where models CLIP and CLIP+ITM as well as the CLIP+SimCLR models trained without the MLM loss start to surpass models trained with MLM in linear probe accuracy.

While for models trained on the full dataset, we observed that CLIP+MLM slightly outperforms the other models on linear probes, we see here that there still remains an approximately 5% gap in accuracy at the 40% fraction. We can only speculate again based on our prior observations that the linear probe accuracy for MLM models, particularly for CLIP+MLM, will continue to increase until it matches and possibly surpasses those of other models.

This data suggests that unlike in the zero-shot tasks, where we observe the trend of models trained with MLM outperforming models trained without the additional language loss, that the benefit of self-supervision in the language domain on visual representations only starts to accrue with growing size of the training dataset. Since the zero-shot tasks are as dependent on the quality of the text representations as on the visual representations, especially on their alignment in the multimodal embedding space, we can observe there that the benefit of training with MLM starts to come earlier than in linear probing. The additional self-supervised language loss in that context makes sense: masked language modelling makes the text representations more robust and thus benefits performance in zero-shot tasks.

The effect of the language loss on visual representations on the other hand is only indirect, through better alignment of representations in the contrastive space. This improves the quality of the visual representations in the image encoder only after the fact, and probably with more data. But since the linear probing task only evaluates the quality of these visual representations and not the quality of the learned contrastive space, we generally see less positive impact in this task than in the zero-shot tasks.

ITM AS POTENTIAL REGULARIZER IN LOW-DATA SETTING As an additional finding, we observe that at the 10% data fraction model CLIP+SimCLR+ITM achieves an almost 5% absolute improvement in accuracy on the linear probes. Since this gap in performance is closed at the 20% fraction and does not establish itself afterwards, it throws up the question of what the cause for this outlier is. A tentative hypothesis is that the ITM loss might act as a sort of regularizer at this specific fraction of training data. We have observed in practice that training models in this low-data regime requires a much lower learning rate than at higher fractions of the dataset. The standard learning rate that worked for models on the full data scale and in the middle fraction range was 0.0002. Because this learning rate would lead to a stuck loss value in the low fractions, we decreased the learning rate by a factor of 10 to 0.00002 at which the loss was able to decrease. The CLIP+SimCLR+ITM model was unique in this case as we were able to successfully train it with the usual learning rate of 0.0002. But in practice the ITM loss overfit to a large degree. So while the ITM head itself was not able to learn any useful features out of the data, we speculate that it acts as a sort of regularizer in that instance.

LINEAR PROBING ACROSS LAYERS

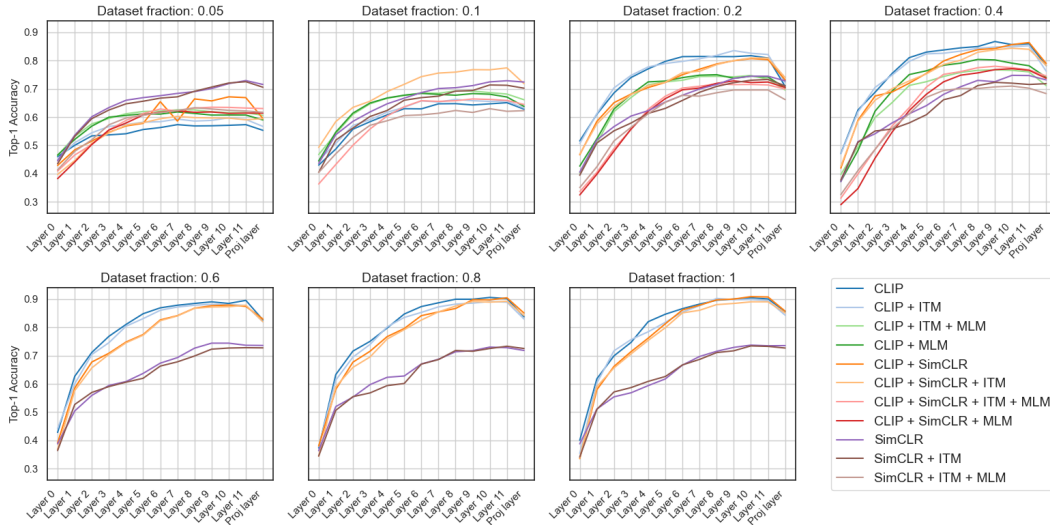


Figure 4.10: Linear probe: Top-1 accuracy per visual encoder layer for Caltech101. Subplots for each fraction of dataset.

Lastly, we can also evaluate how representations across the layers change as we vary the amount of training data. For this, we will look at linear probe accuracies on the Caltech101 dataset. We can visualize the results in two complimentary ways, which will reveal different

insights into the representational structure. Figure 4.10 plots the Top-1 linear probe accuracies at each dataset fraction in individual subplots. As expected and seen before in the evaluation of layer accuracies for the full dataset models, accuracy generally increases with each passing layer, at each dataset fraction. At the 5% fraction, the image-only SimCLR models still outperform the CLIP models. At this fraction of training data, CLIP is simply not exposed to enough training data to learn useful visual representations. Adding SimCLR to CLIP doesn't mitigate this effect in our setting. Starting at 10% of the training set fraction, we see that the CLIP models start to increase in accuracy, and start to match and surpass the SimCLR model from the 20% fraction onwards. As we have seen in the previous section, CLIP+MLM models don't match the performance of CLIP models without MLM even at 40% of the training data. CLIP+SimCLR have similar accuracies to pure CLIP models at the last layers, but see a couple percent gap in accuracy in the middle layers which decreases as we increase the fraction of training data.

Complimentary to this view of the data, we can also plot accuracies for each layer separately, as shown in Figure 4.11. This view reveals different insights into the representations learned by the encoder at each training fraction.

At layer 0, we see a decrease in accuracy for all models as training data grows, as this layer is able to specialize more for low-level features. We are also able to see a big jump in accuracy for models CLIP and CLIP+ITM (shown in blue) going from the 10% to the 20% fraction. At this stage, we have observed in practice that training for the CLIP models stabilizes, and we can train them with higher learning rates. Between layers 1 and 3, increasing the training data does not lead to better accuracy after we increase to more than 20%, as these layers also still specialize in lower-level features. Starting from layer 4, we see an increase in accuracy as we make more training data available.

As observed in the section before, SimCLR models don't benefit from more training data in the later layers.

Models trained with the MLM loss see this decrease in accuracy even in layer 1 and 2, which could be a consequence of these models having a weaker fit to the training data as they have been trained with the same relatively low learning rate as the low fraction models.

4.5 MODEL SIMILARITY ANALYSIS

In this section, we will briefly discuss results of a similarity analysis between representations learned by the different models using the Centered Kernel Alignment.

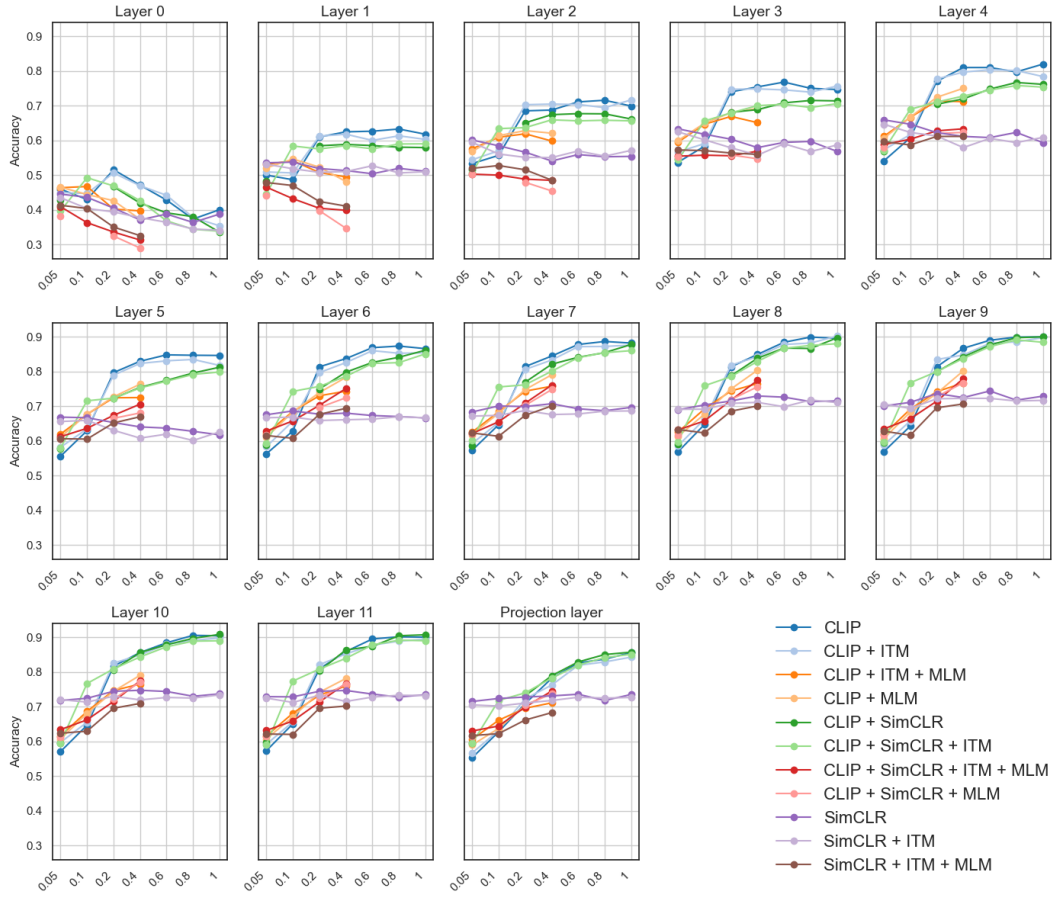


Figure 4.11: Linear probe: Top-1 Accuracy for each image encoder layer vs. fraction of training data.
Dataset: Caltech-101

4 Results

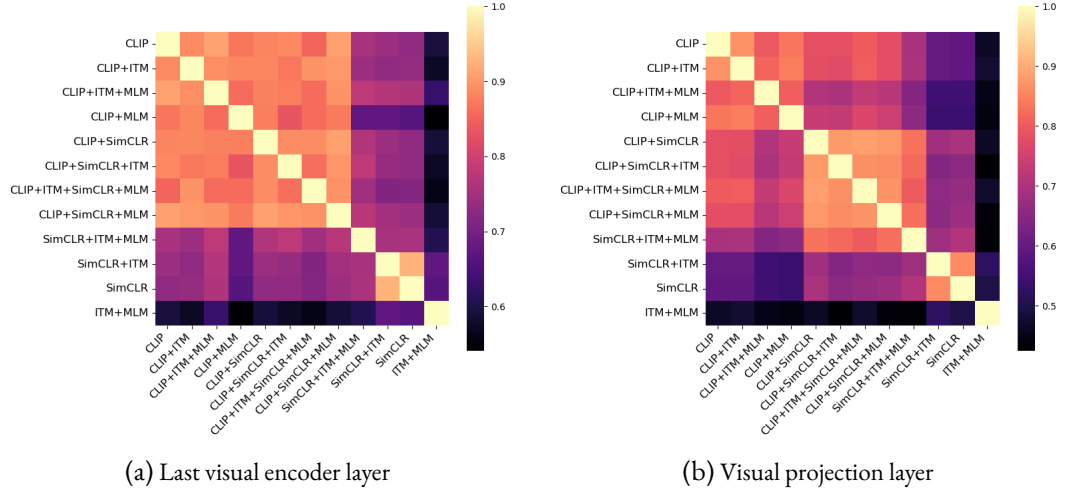


Figure 4.12: CKA scores between model representations, for different image encoder layers. Lighter patches indicate higher similarity. Dataset: CIFAR-10

CENTERED KERNEL ALIGNMENT (CKA)

Figure 4.12 shows CKA scores between different models, for representations of images in the CIFAR-10 test set. We compare representations for two different layers, the last layer of the image encoder and the subsequent projection layer.

On first glance, we can see a shift in representational similarity going from the last encoder layer to the projection layer.

On the last encoder layer, models trained with CLIP form one big cluster with little difference in similarity between them. The CLIP loss dominates the representation space and the other losses don't contribute in a measurable way. Models SimCLR+ITM+MLM, SimCLR+ITM and SimCLR show some degree of similarity with CLIP models, while model ITM+MLM shows little similarity with any of the models. SimCLR+ITM+MLM shows little similarity with SimCLR+ITM and SimCLR, while they show high similarity with each other. This suggests that the addition of the MLM loss has some effect on the representations learned by the image encoder. Model ITM+MLM shows low similarity with any model, no matter if they also utilized the ITM or MLM loss, suggesting that ITM doesn't have a significant influence on the learned representations.

The similarity between models starts to shift on the projection layer, showing more influence from SimCLR. CLIP models don't form as strong of a cluster anymore, and models trained with CLIP+SimCLR start to group together in the middle patch of the plot. Models SimCLR+ITM+MLM also becomes part of that group, to a stronger degree than models SimCLR+ITM and SimCLR, whose similarity to the CLIP+SimCLR models didn't change

much compared to the last layer. SimCLR and SimCLR+ITM also show decreased similarity to CLIP models without SimCLR.

Together, these results show that the SimCLR loss function has an outsized effect on the representation space at the projection layer, as it is applied at that layer.

4.6 DISCUSSION

After our evaluation of the individual downstream tasks, this section summarizes the effects we have observed for each loss function and discusses the implications. Table 4.5 summarizes the effects of each loss on each task.

Table 4.5: Effect of each loss function on downstream tasks. + indicates a positive effect, - indicates a negative effect, and x indicates conditional improvement (here: ITM helps zero-shot classification only with MLM).

Loss Function	Zero-shot Classification	Retrieval	Linear Probe
CLIP	+	+	+
MLM	+	+	+
ITM	x	-	-
SimCLR	-	-	-

CONTRASTIVE IMAGE-TEXT (CLIP) LOSS

The CLIP loss was the dominant loss function throughout all our experiments. Only training with CLIP provided a strong baseline, with good performance on every task. The addition of other loss functions proved either detrimental (ITM, SimCLR) or improved performance by a couple percent (MLM). As seen in the linear probe task, language supervision provided by the CLIP loss produced better representations than self-supervision on images alone (SimCLR). SimCLR showed better performance in the low-data regime, as CLIP was not able to learn good representations from limited data. This finding is in line with results published in related literature. [SDT⁺] We add to this finding by establishing that combining CLIP and SimCLR showed a synergistic effect and lead to better linear probe accuracy in a low-data regime than training with CLIP or SimCLR alone. Representation analysis with CKA substantiates this dominant effect of CLIP, showing that it has the most influence in shaping the visual representation space learned by the image encoder, while being slightly overtaken by SimCLR on the visual projection layer, as the two losses are similarly applied there.

To conclude, CLIP is able to learn strong visual and multimodal representations on its own, and the other loss functions in our experiments served an auxiliary function, with a bit of improvement provided by the unimodal MLM loss.

MASKED LANGUAGE MODELLING LOSS

Our experiments showed a benefit in all downstream tasks when adding MLM to CLIP. This benefit is especially apparent in the zero-shot tasks.

In zero-shot classification, models CLIP+MLM and CLIP+MLM+ITM improve accuracy over CLIP. The latter model is also the only model in our study that didn't suffer the decrease in task performance when embedding the class labels in a template compared to using no template. CLIP+MLM models with added SimCLR or ITM didn't improve accuracy over CLIP, as these loss functions degraded representation quality. In zero-shot retrieval, CLIP+MLM models always improved performance over CLIP, regardless of added SimCLR or ITM. In linear probing, CLIP+MLM was the only model that didn't drop below CLIP in accuracy, with a minimal 1% improvement in accuracy. We can thus state that training CLIP with the additional language masking loss generally improves the models representational quality, even in its unimodal formulation without being conditioned on visual features.

Another supporting factor is that CLIP+MLM models were trained with only half the batch size of non-MLM models. Since research has shown that contrastive losses benefit strongly from larger batch sizes [CKNH], as the increased number of negative examples in a batch facilitates better discrimination in the embedding space, the benefit of additional MLM is even more pronounced.

This benefit naturally accrues more dominantly in the zero-shot tasks, as these tasks are dependent on both the quality of the visual and text representations. The benefit on the purely visual representations learned by the image encoder before projection into the multimodal representation space, on the other hand, is less significant. Still, it seems that the improved representational quality in the language stream also provides small benefits to the visual encoder, and is facilitated by the multimodal CLIP loss. This finding is in line with related literature, which found that fine-tuning a text encoder to read out visual representations in a CLIP-like dual-stream model can improve performance on zero-shot tasks. [ZWM⁺] This suggests that training with CLIP alone leads to strong, generalizable visual representations, and that additional training of the text encoder can lead to more improvements in downstream task performance than additional training of the visual encoder.

IMAGE-TEXT MATCHING LOSS

In most cases, adding ITM to the loss configuration had no beneficial impact on task performance.

ITM decreased accuracy in zero-shot classification when added to any model, and always worsened performance compared to CLIP. The only notable exception is CLIP+MLM+ITM, which is the best performing model on that task, and exhibits special behaviour, as it didn't suffer the degradation in accuracy when going from zero templates to one template for class name embedding. ITM also decreased performance in zero-shot retrieval when added to any loss combination. On the linear probing task, all CLIP+ITM models showed worse performance than CLIP.

SINGLE ITM HEAD IS NOT EXPRESSIVE ENOUGH While not as detrimental as SimCLR, the addition of ITM in our model setup did not prove beneficial. Other works that utilize an Image-Text Matching loss use more extensive cross-modal interaction layers. For example, ALBEF [LSG⁺] takes the representations of the unimodal encoders and processes them jointly with a multimodal encoder using cross-attention for six layers before applying the ITM loss. Other dual-stream designs like ViLBERT [LBPL] also employ cross-attention layers to capture multimodal interactions for ITM, and single-stream models like ViLT [KSK] use regular self-attention layers to extensively model the cross-modal interactions. In contrast, the ITM head in our design only consists of a two-layer MLP that takes the concatenated 512-dim embeddings of the last layer as input. This head on top of the unimodal encoders doesn't seem to be able to capture the fine-grained interactions between image and text that are necessary to produce good multimodal representations.

SIMCLR LOSS

On every tested downstream task, from zero-shot classification to image-text retrieval to linear probing, adding SimCLR proved detrimental. Adding it to any configuration of losses further decreased performance on all tasks. CLIP+SimCLR models exhibited worse performance than CLIP in zero-shot classification and linear probing. In retrieval, CLIP+SimCLR models trained with MLM showed better performance than CLIP, but this is more likely attributed to the positive effect of the MLM loss.

CLIP AND SIMCLR INTERFERE AT THE PROJECTION LAYER The application of SimCLR and CLIP to the outputs of the same layer is the likely cause of the observed degradation in performance. One could hypothesize that applying SimCLR and CLIP to the same projec-

tion layer would lead to more robust multimodal embeddings that are invariant to the transformations applied to the images. But the decrease in task performance when using CLIP and SimCLR together are evidence that it leads to worse multimodal and visual representations, as seen in zero-shot and linear probe performance. While the authors of SimCLR found that the representations learned by the layer before the SimCLR projection head perform better in downstream tasks than the representations from the projection layer [CKNH], adding SimCLR to CLIP does not lead to better representations on that layer than just training with CLIP. The interference between the loss functions is thus not able to confer the benefits that training with SimCLR could provide.

A similar work to this, SLIP [MKWX], uses different projection heads for SimCLR and CLIP losses, and reports improvements in downstream task performance compared to only training with CLIP. We thus encourage future research to use our experimental setup, with a change in SimCLR implementation.

STRONGER AUGMENTATIONS LEAD TO MORE INTERFERENCE

Our experiments in training models with different augmentation strengths allow us to investigate the effect that different augmentation settings have on the CLIP+SimCLR models in our setup. Figure 4.16 plots performance for models CLIP, CLIP+ITM, CLIP+SimCLR and CLIP+SimCLR+ITM on all three downstream tasks for different augmentation strengths, as well as SimCLR and SimCLR+ITM on linear probing.

In linear probing, we see a general increase in accuracy on a model-by-model basis with stronger augmentations. Models CLIP, CLIP+ITM and SimCLR see a 5% increase in accuracy, while model SimCLR+ITM benefits slightly less. CLIP+SimCLR models see a modest improvement of 2-3%. While increased augmentations improve task performance, we also observe that the relative accuracy between CLIP without SimCLR and CLIP with SimCLR models changes as augmentation strength increases. In the weaker augmentation setting, CLIP+SimCLR models show comparable, if not slightly better performance than pure CLIP. The relative performance changes with stronger augmentations, as CLIP+SimCLR models become less accurate than CLIP. So while training with stronger augmentations leads to better performance for every single model, they also lead to more interference between the CLIP and SimCLR losses, and CLIP+SimCLR models see less improvement than CLIP models trained without SimCLR.

The same pattern is seen in zero-shot retrieval. CLIP+SimCLR models show better performance than CLIP and CLIP+ITM in a weaker augmentation setting, but start to perform worse when increasing strength of the augmentations.

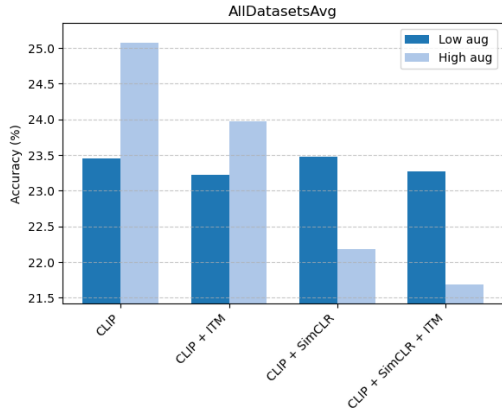


Figure 4.13: Zero-shot classification (Top-1 accuracy)

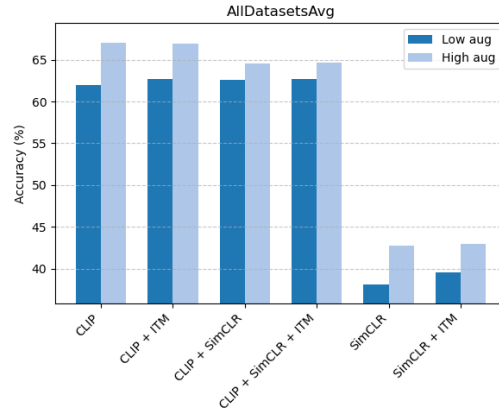


Figure 4.14: Linear probing (Top-1 accuracy)

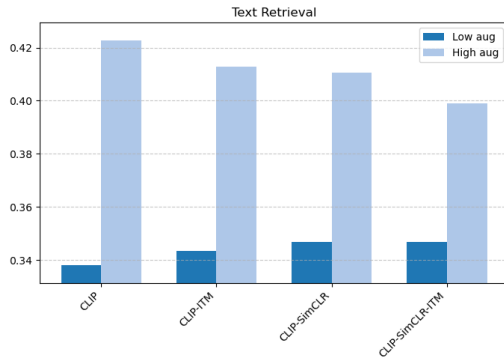
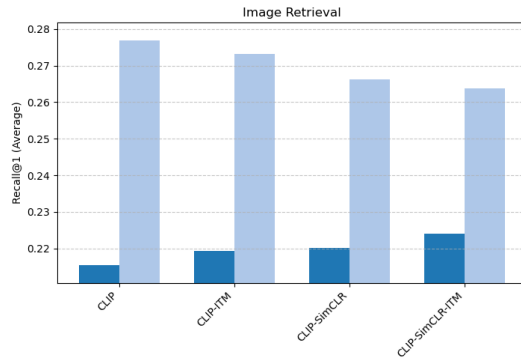


Figure 4.15: Zero-shot retrieval (Image and Text) (Recall@1)

Figure 4.16: Comparison of models with losses CLIP and SimCLR on all three downstream tasks for different image augmentation strengths during model training.

4 Results

Contrary to the other tasks, in zero-shot classification increasing augmentation strength doesn't lead to improved performance for models CLIP+SimCLR and CLIP+SimCLR+ITM, as accuracy for both models drops by about 1.5%.

5 CONCLUSION

We end this thesis with a summary and discussion on limitations and directions for future research.

5.1 SUMMARY

This thesis investigated the influence of different loss functions in multimodal learning on downstream tasks, in the context of vision-language models.

By fixing the model architecture and all other training parameters, we were able to ablate the choice of loss functions utilized in model training to isolate the effects of each individual loss, alone and in combination. Specifically, we chose a simple dual-stream encoder design, with a Vision Transformer as the image encoder and a BERT Transformer as the text encoder. For our choice of loss functions, we selected a contrastive image-text loss (CLIP) and an image-text matching loss (ITM) as our multimodal losses, and one unimodal loss for each modality: a contrastive image loss (SimCLR) for the image encoder and a masked language modelling loss (MLM) for the text encoder. The unimodal losses operated only on their respective streams, without being conditioned on the other modality. The multimodal interaction was kept intentionally simple, and was restricted to only one linear projection layer at the end of the respective encoders for the CLIP loss, and to the projection layer followed by a two-layer MLP matching head taking a concatenation of the embeddings of the two modalities for the ITM loss.

We trained a selection of 12 models with different combinations of our selected loss functions, and evaluated their performance on three downstream tasks after training: zero-shot image classification, zero-shot image-text retrieval and linear probe image classification on vision encoder representations. We also analyzed similarities between model representation using Centered Kernel Alignment (CKA).

We pre-trained each model on a combined dataset of image-caption pairs (Conceptual Captions 3M + COCO Captions + Visual Genome) spanning 3.95M images. We also performed an extensive ablation study on the size of the training dataset, training different model

combinations with fractions of the full training dataset ranging from 5% to 100%.

Our experiments showed that the CLIP loss is the most powerful of all investigated loss functions, producing performant models with generalizable visual representations and good capabilities in zero-shot classification and image-text retrieval. Models trained with Image-Text Matching were not able to learn better representations than models trained with CLIP. We showed that the language supervision facilitated by the CLIP loss produced better representations for downstream transfer than self-supervision on images alone, and that the addition of unimodal losses can improve learning in a low-data regime. While our implementation of SimCLR lead to a degradation in performance, the addition of Masked Language Modelling benefitted CLIP on all downstream tasks.

5.2 LIMITATIONS AND FUTURE RESEARCH

ARCHITECTURAL LIMITATIONS In order to properly isolate the effect that loss functions have on model representations and downstream performance, it is a necessity to fix every other variable in model training, including model architecture. Our selection of the CLIP loss for this study necessitated a dual-stream encoder design without cross-modal interactions before the final projection layer, as they would interfere with the contrastive loss function. This in turn precluded the usage of masking losses that are conditioned on features of the other modality. It also motivated the specific design of our Image-Text Matching loss, which contrary to many works in the literature [LSG⁺, KSK], was not trained on intermodal tokens that are conditioned on both modalities. This is a necessary trade-off when designing an ablation study on loss functions, as every experimental variable other than the loss function has to be controlled for in order to properly assess the effect of the loss.

A direction for future research is thus to choose a different architecture that can accommodate different loss functions. A single-stream architecture does not allow for a contrastive image-text loss, but enables cross-modal masking losses and an Image-Text Matching loss that operates on multimodal representations. Similarly, a dual-stream design with cross-modal attention layers also allows for conditional losses and more extensive ITM.

SELECTION OF LOSS FUNCTIONS Even though our chosen architecture has its limitations, there are still loss functions that can be added too it without interfering with the other losses. We thus encourage future research to use our experimental design to add different loss functions as an extension of the ablation study. One simple extension that is symmetrical to the text encoder would be the addition of Masked Patch Prediction on the image encoder, where

the task is to predict visual patches that were masked as input. This loss could prove fruitful in improving visual representation quality for downstream tasks.

Another addition could be a fine-grained contrastive loss calculated between all token embeddings of the modalities instead of contrasting only the global representation token, as proposed in FILIP [YHH⁺]. This loss function could enhance learning of fine-grained multimodal features and enhance zero-shot, particularly in image-text retrieval.

LIMITED EVALUATION OF IMAGE-TEXT MATCHING Another limitation of this study is that we didn't thoroughly investigate the performance of models trained with ITM as the only multimodal loss. Specifically in the zero-shot tasks, we only evaluate performance of models trained with CLIP and other loss functions added in an auxiliary function. Our only method for obtaining predictions in the zero-shot setting is by computing the similarity scores between the normalized embeddings of the projection layer, which was trained by CLIP. For models with ITM, we tested the usefulness of the matching head by pre-selecting predictions through cosine similarity and obtaining the final prediction with the ITM head.

Another approach would be to use the ITM head directly for zero-shot prediction, by constructing concatenated pairs of image embeddings and class label embeddings (in zero-shot classification) or caption embeddings (in zero-shot retrieval) obtained from the projection layer and computing matching scores for each pair with the ITM head, selecting the class/image or caption with the highest matching score. With our chosen approach, we were only able to assess the usefulness of ITM for zero-shot tasks together with CLIP.

As a direction for future research, models trained with ITM could be evaluated for zero-shot tasks by using the matching head instead of the cosine similarity between the image and text embeddings. Models trained with CLIP+ITM could be evaluated with both methods. This approach would enable a thorough view of the usefulness of each loss function in zero-shot performance, although our preliminary results in zero-shot classification and linear probing suggest that our ITM implementation is not expressive enough to fully capture the multimodal interactions.

IMPLEMENTATION OF SIMCLR LOSS Another direction for future research could be to implement the SimCLR loss as proposed in SLIP [MKWX], which implements a SimCLR loss on top of a CLIP loss, with a similar dual-stream model design as in this study, but uses a separate projection head for application of SimCLR instead of the same projection layer for both losses, as was done in this work. SLIP reports improvements in downstream task performance over the standard CLIP approach, while our study showed a decrease in performance, where it seems that optimization of the two objectives on outputs of the same layer

5 Conclusion

leads to interference and reduces representation quality. Future works could thus use our study design and only change the implementation of SimCLR by adding a separate head to the model architecture.

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SUPPLEMENTARY TABLES

Table 1: Zero-shot classification: Top-1 accuracies (%) for text classifier with no prompt template.

Model	CIFAR10	CIFAR100	Caltech101	Caltech256	DTD	FGVCaircraft	Food101	ImageNet	OxfordPets	Places365	STL10	StanfordCars	All Avg	General Avg	Fine Avg
CLIP	52.01	19.98	43.45	33.76	14.63	1.20	15.05	15.04	10.38	16.50	81.56	0.82	25.37	40.97	9.76
CLIP + ITM	50.65	20.72	44.01	33.20	13.56	1.02	14.44	14.92	7.91	17.20	84.18	0.80	25.22	41.28	9.15
CLIP + SimCLR	48.82	18.55	36.64	30.35	14.68	1.17	15.08	14.75	8.40	16.04	77.60	0.77	23.57	37.78	9.36
CLIP + SimCLR + ITM	42.96	18.73	44.06	31.37	16.76	1.44	13.81	14.22	8.75	15.79	79.08	0.79	23.98	38.40	9.55
CLIP + MLM	45.66	19.38	46.61	36.43	17.13	0.78	14.52	16.37	6.44	19.87	83.22	0.86	25.61	41.28	9.93
CLIP + ITM + MLM	51.72	20.28	43.45	36.93	11.81	1.14	13.40	16.19	9.62	19.23	80.80	1.12	25.47	41.56	9.39
CLIP + SimCLR + MLM	45.45	20.64	42.62	34.35	16.01	0.93	14.89	15.65	9.65	18.89	79.42	0.90	24.95	39.69	10.21
CLIP + ITM + SimCLR + MLM	43.24	22.26	43.58	34.70	16.54	0.90	15.37	16.07	9.81	18.45	76.18	0.95	24.84	39.34	10.34
SimCLR	9.98	1.25	0.68	0.32	3.03	0.60	1.05	0.11	2.66	0.23	9.82	0.41	2.51	3.69	1.33
SimCLR + ITM	10.33	0.68	0.72	0.41	1.97	0.75	1.13	0.13	2.23	0.25	10.40	0.47	2.45	3.78	1.13
SimCLR + ITM + MLM	12.98	0.90	3.77	0.82	1.44	1.05	0.76	0.11	3.59	0.20	10.90	0.37	3.07	4.91	1.23
ITM + MLM	7.57	1.09	0.59	0.28	2.23	0.90	1.04	0.04	2.99	0.28	3.18	0.69	1.74	2.13	1.35

Table 2: Zero-shot classification: Top-1 accuracies (%) for text classifier with 1 prompt template.

Model	CIFAR10	CIFAR100	Caltech101	Caltech256	DTD	FGVC	Food101	ImageNet	OxfordPets	Places365	STL10	StanfordCars	All Avg	General Avg	Fine Avg
CLIP	48.14	14.30	40.08	32.07	13.94	1.50	13.66	11.58	7.91	14.51	70.70	0.82	22.43	36.14	8.72
CLIP + ITM	42.24	17.21	39.26	30.53	17.87	1.14	13.04	11.48	6.09	13.82	68.08	0.79	21.80	34.80	8.79
CLIP + SimCLR	41.96	11.29	33.87	27.90	15.96	1.11	12.24	10.31	7.64	13.28	63.52	0.55	19.97	31.48	8.46
CLIP + SimCLR + ITM	34.34	10.21	32.10	26.60	14.41	1.23	11.82	10.16	7.77	14.12	66.86	0.56	19.18	30.04	8.32
CLIP + MLM	44.64	18.08	48.83	35.95	15.37	1.29	13.51	12.43	7.72	17.89	73.10	0.85	24.14	38.84	9.44
CLIP + ITM + MLM	58.13	21.18	47.82	37.86	15.05	1.35	13.04	13.44	8.21	19.15	78.08	0.76	26.17	42.75	9.59
CLIP + SimCLR + MLM	44.33	19.55	41.64	34.34	17.45	0.84	13.22	11.63	8.99	17.44	70.28	0.75	23.37	36.96	9.78
CLIP + ITM + SimCLR + MLM	46.34	10.88	40.05	30.26	18.03	0.90	12.97	11.29	8.78	17.43	70.40	0.63	22.33	34.87	9.79
SimCLR	10.00	1.31	0.75	0.25	2.39	0.81	1.01	0.13	2.83	0.29	9.78	0.45	2.50	3.70	1.30
SimCLR + ITM	10.02	0.84	1.51	0.46	2.39	1.05	1.05	0.09	2.66	0.22	10.24	0.71	2.60	3.86	1.35
SimCLR + ITM + MLM	12.51	0.85	0.59	0.37	1.65	1.50	0.77	0.07	2.64	0.20	10.34	0.27	2.65	4.12	1.17
ITM + MLM	8.40	0.91	0.49	0.12	1.60	0.69	0.86	0.06	1.71	0.23	3.12	0.45	1.55	2.18	0.92

Table 3: Zero-shot classification: Top-1 accuracies (%) for text classifier with 3 prompt templates.

Model	CIFAR10	CIFAR100	Caltech101	Caltech256	DTD	FGVC	Food101	ImageNet	OxfordPets	Places365	STL10	StanfordCars	All Avg	General Avg	Fine Avg
CLIP	55.47	18.39	41.95	34.43	15.43	1.32	14.26	13.51	7.23	17.29	80.28	0.79	25.03	40.67	9.38
CLIP + ITM	47.48	18.96	37.83	32.36	18.16	1.24	12.95	12.64	8.25	16.45	75.83	0.76	22.99	36.55	9.44
CLIP + SimCLR	40.39	14.22	37.29	30.56	17.13	1.11	13.25	11.79	8.67	14.64	72.44	0.79	21.85	34.45	9.26
CLIP + SimCLR + ITM	43.41	11.73	35.10	26.57	14.15	1.44	11.64	10.91	8.33	14.75	70.59	0.83	20.21	32.24	8.18
CLIP + MLM	47.49	20.43	47.08	37.16	15.64	0.87	14.02	13.44	7.91	18.82	77.64	0.65	25.10	40.54	9.65
CLIP + ITM + MLM	54.79	20.55	46.23	37.30	15.21	1.14	12.91	13.24	7.99	18.41	81.20	0.92	26.04	42.56	9.52
CLIP + SimCLR + MLM	46.87	21.44	43.38	35.68	18.56	0.69	13.55	12.81	9.95	18.75	76.00	0.81	24.87	39.36	10.39
CLIP + ITM + SimCLR + MLM	46.08	13.19	37.89	29.64	16.76	0.96	11.48	11.20	8.26	17.10	74.20	0.82	22.55	35.76	9.35
SimCLR	10.04	1.19	0.66	0.29	1.81	0.66	1.03	0.10	2.85	0.33	8.28	0.43	2.31	3.43	1.19
SimCLR + ITM	9.72	0.98	1.47	0.39	2.26	0.99	0.10	2.73	0.32	9.68	0.50	2.57	3.79	3.79	1.34
SimCLR + ITM + MLM	13.78	3.69	5.39	1.82	4.26	1.32	1.57	0.30	4.08	1.19	11.47	0.53	2.73	4.66	1.98
ITM + MLM	11.04	1.37	1.02	0.45	2.93	0.96	1.24	0.10	2.34	0.48	4.42	0.51	1.82	2.78	1.65

Table 4: Zero-shot classification: Top-1 accuracies (%) for text classifier with 5 prompt templates.

Model	CIFAR10	CIFAR100	Caltech101	Caltech256	DTD	FGVC	Food101	ImageNet	OxfordPets	Places365	STL10	StanfordCars	All Avg	General Avg	Fine Avg
CLIP	55.00	17.86	44.04	35.35	15.90	1.23	14.48	13.58	6.98	17.24	80.72	0.74	25.26	41.09	9.43
CLIP + ITM	45.07	18.98	38.96	33.04	18.03	1.30	12.99	12.52	8.74	17.10	76.66	0.82	23.00	36.39	9.61
CLIP + SimCLR	41.84	14.60	37.75	31.01	17.61	1.05	13.46	11.30	7.88	14.59	74.18	0.85	22.18	35.11	9.24
CLIP + SimCLR + ITM	44.17	11.70	35.84	27.33	14.57	1.38	11.80	11.01	9.31	15.36	70.68	0.86	20.52	32.49	8.56
CLIP + MLM	47.75	21.26	47.98	37.47	15.74	0.99	14.64	13.39	7.09	18.73	78.16	0.66	25.32	41.00	9.64
CLIP + ITM + MLM	56.46	20.59	45.97	37.73	15.05	1.17	13.21	13.16	7.61	18.59	80.54	0.82	26.12	42.74	9.51
CLIP + SimCLR + MLM	47.72	20.49	44.06	36.40	18.78	0.90	13.73	12.85	9.65	19.41	76.40	0.65	25.09	39.65	10.52
CLIP + ITM + SimCLR + MLM	48.11	15.70	39.13	30.76	16.33	0.99	12.63	11.12	8.07	17.14	76.54	0.98	23.39	37.31	9.47
SimCLR	10.10	1.13	0.59	0.33	2.02	0.90	1.03	0.09	2.80	0.33	7.98	0.42	2.31	3.37	1.25
SimCLR + ITM	10.09	1.00	1.92	0.40	1.94	0.82	0.91	0.08	2.88	0.29	9.96	0.53	2.67	4.07	1.27
SimCLR + ITM + MLM	13.22	3.68	5.85	1.89	4.15	1.41	1.60	0.33	3.70	1.16	12.09	0.47	3.01	4.84	1.92
ITM + MLM	11.27	1.38	1.02	0.44	2.85	0.85	1.25	0.11	2.26	0.48	5.83	0.48	1.93	3.02	1.61

Table 5: Zero-shot classification: Top-1 accuracies (%) for text classifier with 9 prompt templates.

Model	CIFAR10	CIFAR100	Caltech101	Caltech256	DTD	FGVC	Food101	ImageNet	OxfordPets	Places365	STL10	StanfordCars	All Avg	General Avg	Fine Avg
CLIP	55.83	20.80	42.56	35.96	17.61	1.14	15.30	15.40	9.38	18.66	82.90	0.86	26.37	42.24	10.49
CLIP + ITM	49.96	19.87	40.57	33.94	18.70	1.18	14.11	14.05	9.71	18.04	79.93	0.82	24.34	38.54	10.14
CLIP + SimCLR	37.18	15.22	37.98	31.98	19.20	1.11	14.42	13.11	8.80	16.19	77.80	0.90	22.82	35.54	10.10
CLIP + SimCLR + ITM	46.17	13.57	38.89	29.36	15.98	1.38	12.71	12.69	9.06	16.03	74.36	0.81	21.84	34.77	8.91
CLIP + MLM	49.85	21.76	49.29	38.89	16.54	1.02	15.49	15.37	8.37	20.02	82.16	0.74	26.62	42.89	10.36
CLIP + ITM + MLM	57.72	22.00	46.20	38.07	14.68	0.84	13.10	14.37	9.05	19.06	79.78	0.77	26.50	43.28	9.71
CLIP + SimCLR + MLM	50.39	21.66	43.64	37.46	19.26	0.72	14.43	14.55	10.41	19.72	79.10	0.88	26.02	41.13	10.90
CLIP + ITM + SimCLR + MLM	50.74	18.47	39.65	32.53	17.71	1.05	13.06	12.63	8.83	17.67	78.88	0.88	24.65	39.25	10.05
SimCLR	10.04	1.23	0.58	0.23	1.81	0.72	1.03	0.10	2.93	0.33	7.80	0.43	2.27	3.33	1.21
SimCLR + ITM	9.78	0.95	0.52	0.34	3.01	1.05	0.98	0.11	2.72	0.32	9.90	0.49	2.54	3.69	1.40
SimCLR + ITM + MLM	13.17	3.63	5.85	1.98	4.26	1.20	1.49	0.34	4.46	1.13	14.73	0.48	3.48	4.29	1.99
ITM + MLM	11.70	1.37	0.94	0.34	2.85	0.78	1.27	0.10	2.15	0.50	10.66	0.50	2.31	3.35	1.53

Table 6: Zero-shot classification: Top-1 accuracies (%) for text classifier with 18 prompt templates.

Model	CIFAR10	CIFAR100	Caltech101	Caltech256	DTD	FGVC	Food101	ImageNet	OxfordPets	Places365	STL10	StanfordCars	All Avg	General Avg	Fine Avg
CLIP	53.95	19.95	41.92	35.46	17.61	1.05	15.22	15.41	9.27	19.13	82.32	0.76	26.00	41.50	10.51
CLIP + ITM	46.95	19.47	39.62	33.31	17.47	1.26	14.03	13.93	8.98	18.42	79.61	0.79	23.75	37.64	9.86
CLIP + SimCLR	37.60	15.27	37.07	31.71	18.46	1.17	14.57	13.03	8.56	16.44	77.38	0.95	22.68	35.34	10.02
CLIP + SimCLR + ITM	45.85	13.14	38.48	29.49	15.45	1.26	12.53	12.55	9.05	16.14	74.56	0.82	21.69	34.62	8.77
CLIP + MLM	48.12	22.10	49.72	39.02	16.01	1.02	15.62	15.22	8.10	20.45	82.12	0.81	26.53	42.72	10.34
CLIP + ITM + MLM	56.38	21.94	46.23	38.22	15.05	0.90	13.25	14.30	9.10	19.92	79.02	0.88	26.48	42.97	10.00
CLIP + SimCLR + MLM	48.47	22.06	44.71	37.57	18.40	0.72	14.60	14.43	10.05	20.40	79.60	0.79	25.98	41.14	10.83
CLIP + ITM + SimCLR + MLM	49.35	18.14	39.40	31.81	17.18	0.78	12.90	12.13	8.61	18.23	79.26	0.82	24.31	38.71	9.91
SimCLR	10.10	1.26	0.61	0.23	1.81	0.78	1.01	0.10	2.93	0.32	7.88	0.43	2.29	3.36	1.21
SimCLR + ITM	9.65	0.97	0.88	0.32	2.26	0.96	1.06	0.10	2.47	0.22	10.65	0.47	2.46	3.75	1.18
SimCLR + ITM + MLM	11.81	3.56	7.94	2.21	3.72	1.11	1.42	0.30	4.05	1.23	14.09	0.43	3.01	4.01	1.86
ITM + MLM	11.36	1.34	1.16	0.41	3.27	0.81	1.24	0.09	2.46	0.52	10.42	0.47	2.24	2.98	1.67

Model	CIFAR10	CIFAR100	Caltech101	Caltech256	DTD	FGVCAircraft	Food101	ImageNet-100-0.01	ImageNet-100-0.1	OxfordIIITPet	Places365	STL10	StanfordCars	Average (All)	Average (General object)	Average (Fine-grained)
CLIP	87.43	69.34	88.19	77.90	68.78	33.96	67.79	62.46	74.32	67.62	47.13	92.95	33.84	67.05	78.94	53.19
CLIP + ITM	87.64	69.00	89.80	77.10	67.66	34.65	67.28	61.34	74.84	67.76	46.93	92.94	33.54	66.96	78.95	52.97
CLIP + SimCLR	85.55	66.27	86.52	76.28	67.87	30.18	67.49	58.34	72.48	64.70	45.75	91.14	27.11	64.59	76.65	50.52
CLIP + SimCLR + ITM	85.52	66.26	88.65	75.50	68.14	30.72	66.84	57.54	73.32	62.69	45.91	91.45	27.32	64.60	76.89	50.27
CLIP + MLM	88.15	70.10	90.03	79.06	69.15	34.08	67.99	63.02	75.24	68.82	47.80	93.69	34.63	67.83	79.90	53.75
CLIP + ITM + MLM	86.82	68.54	90.15	77.47	67.02	33.78	64.53	61.38	74.06	68.27	46.94	92.90	33.95	66.60	78.76	52.42
CLIP + SimCLR + MLM	86.39	67.84	88.48	76.48	68.30	29.79	66.96	59.60	74.26	65.03	46.30	92.27	28.17	65.38	77.90	50.76
CLIP + ITM + SimCLR + MLM	87.12	68.80	89.06	76.61	68.56	29.16	68.89	60.24	74.56	65.19	46.69	93.01	29.39	65.95	78.49	51.31
SimCLR	71.65	49.79	69.30	47.86	45.64	16.80	38.65	26.82	40.18	33.80	28.95	71.72	14.07	42.71	53.90	29.65
SimCLR + ITM	71.99	50.21	68.84	48.19	45.27	17.88	38.95	27.52	41.06	34.15	29.01	72.09	13.99	43.01	54.27	29.88
SimCLR + ITM + MLM	79.62	58.45	77.48	61.11	59.52	21.30	51.35	40.78	56.90	45.24	37.49	81.00	16.28	52.81	65.05	38.53
ITM + MLM	62.37	37.38	57.89	33.40	34.73	8.46	20.03	18.58	29.60	21.75	21.09	64.88	7.00	32.09	43.44	18.84

Table 7: Linear probing: Top-1 accuracy for every model on each dataset, with average scores for three dataset groups.

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ERKLÄRUNG ZUR ABSCHLUSSARBEIT

gemäß § 34, Abs. 16 der Ordnung für den Masterstudiengang Informatik
vom 17. Juni 2019

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